

# Distributional Uncertainty and Adaptive Decision-Making in System Co-design (Extended Version)

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**Abstract**—Complex engineered systems require coordinated design choices across heterogeneous components under conflicting objectives and uncertain specifications. Monotone co-design provides a compositional framework for such problems. Performance of each subsystem is modeled with a design problem: a relation specifying what resources suffice to provide each functionality. Existing uncertain co-design models rely on interval bounds, which support worst-case reasoning but cannot represent probabilistic risk or multi-stage adaptive decisions. We develop a distributional extension of co-design that models uncertain design outcomes as distributions over design problems and supports adaptive decision processes through Markov-kernel re-parameterizations. Using quasi-measurable and quasi-universal spaces, we show that the standard co-design compositions remain compositional under this richer uncertainty, and introduce queries and observations extracting probabilistic trade-offs, including feasibility probabilities, confidence bounds, and distributions of minimal required resources. A task-driven unmanned aerial vehicle case study shows how the framework captures risk-sensitive and information-dependent design choices that interval models cannot express.

**Index Terms**—Compositional design, Uncertain systems, Adaptive systems, Large-scale systems

## I. INTRODUCTION

Embodied intelligence and cyber-physical systems (CPS) tightly couple heterogeneous hardware (sensors, actuators) with software (perception, planning) and must be evaluated against multiple, often incomparable, metrics, so designers face nontrivial *trade-offs* across components, subsystems, and system-level objectives while coordinating stakeholders with diverse expertise. For safety- and mission-critical applications, uncertainty cannot be treated purely adversarially: designers must quantify *risk* (e.g., probability of meeting specifications) and reason about how robustness trades against performance. Design is moreover rarely one-shot, as decisions are staged while information arrives, so later choices should be *adaptive* to earlier commitments and intermediate observations.

Monotone co-design provides a compositional framework for such design [1]–[3]. Performance of each component is modeled as a design problem (DP): a relation specifying which

resources suffice to provide a functionality, both forming partially ordered sets. System performance is built by interconnecting DPs in series, parallel, and feedback, with traced symmetric monoidal structure [2], enabling joint optimization of hardware/software architectures in robotics and control [4]–[6], transportation [7], and automotive [8].

Existing co-design tools, however, treat uncertainty via interval bounds on feasible sets [2], [9]. Intervals support worst-case robustness but are too coarse for probabilistic risk (chance constraints, confidence levels, quantiles) or multi-stage adaptive policies: they cannot separate “almost-always safe” designs from uniformly conservative ones, nor express strategies reacting to intermediate observations.

This paper extends monotone co-design with *distributional* uncertainty, thereby also enabling *adaptive* decisions. We model uncertain design outcomes via probability measures (and parametric families) over spaces of DPs, with quasi-measurable structures avoiding measurability pathologies while preserving compositionality. Here *distributional* means uncertainty is modeled with distributions supplied to the framework, rather than robust sets or intervals; the same term is sometimes used for ambiguity sets of distributions [10], which is not this paper’s topic though we note it as a future direction. This enables (i) distributionally uncertain trade-offs (design problems) as design results, (ii) multi-stage adaptive design policies conditioning later decisions and observations on earlier realized outcomes, and (iii) queries returning probabilistic trade-offs, uncertain maps, and risk metrics (e.g., expected resource use, quantiles) rather than only deterministic feasible sets, all while preserving measurability. This is the first framework that simultaneously handles general distributional uncertainty over compositional design models, supports multi-stage adaptivity, and remains closed under the standard co-design compositions.

## A. Related work

This work sits at the intersection of system-level design for CPS, compositional specification formalisms, and robust and stochastic optimization. The recurring gap: rich probabilistic and adaptive uncertainty models lack a modular, multiobjective language for heterogeneous subsystems; while compositional frameworks use descriptions too coarse for quantitative risk or staged policies.

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1) *System-level design in robotics and CPS*: Co-optimizing hardware and software at the system level is long recognized as central to robotics and CPS [2], [11], [12], including multi-objective formulations for robot system design [13] and cross-layer CPS methodologies spanning sensing, computation, and control [14], [15]. Recent work also studies cost–performance trade-offs in probabilistic planning [16]. These efforts target specific problem classes; a framework combining heterogeneous components, multiobjective trade-offs, and distributional uncertainty is missing [17].

2) *Contract-based and compositional design*: Assume–guarantee contracts enable modular reasoning by associating components with environment assumptions and behavioral guarantees [18], with applications spanning circuits and autonomous systems [19], [20]. They naturally support open-system composition [21], but require substantial adaptation when subsystems interact through explicit interfaces [18], [22]. Contracts moreover emphasize satisfying specifications rather than trade-offs, and optimization is typically limited to the weakest component contract for a given environment [18]. [23], [24] combine mixed-integer programming with contracts to minimize cost subject to specifications but remain single-objective, and probabilistic extensions target satisfaction probabilities rather than compositional optimization over objectives and risk [25].

3) *Robust, stochastic, and adaptive optimization*: Robust and stochastic optimization offer mature tools for uncertainty, including ambiguity sets, distributional formulations, and chance constraints [26]–[28], though many target a single scalar metric (e.g., expected cost [26], [29]) rather than trade-offs across interconnected subsystems. Chance constraints are addressed by discretization [30] and gradient methods [31], [32], where randomized policies become essential [33]. Multi-stage stochastic programming models *here-and-now* versus *wait-and-see* decisions via policies over evolving uncertainty [27], [34]; general formulations exist [29], [34], but implementations rely on scenario trees [35], [36], restrictive under continuous uncertainty and commonly abstract away how observations depend on prior design choices [26], [28]. We import these distributional and adaptive ideas into co-design.

4) *Monotone co-design and uncertainty*: Monotone co-design models components as monotone relations between functionalities and resources, composed in series/parallel/feedback while preserving order and enabling optimization [2]. It is instantiated across autonomous systems, control, and beyond [4]–[8]. Uncertainty there is treated by interval models guaranteeing conservative feasibility [9], which encode neither probabilistic risk nor adaptive policies.

## B. Statement of Contribution

Our contributions are fourfold. First, we introduce probabilistic measures and parametric uncertainty over DPs using quasi-measurable spaces, a distributional uncertainty model closed under co-design compositions. Second, we formalize multi-stage design processes whose implementations and specifications depend on intermediate observations and earlier choices, generalizing interval uncertainty compositionally. Third, we

define query operators from distributional uncertain design results to probabilistic performance characterizations, enabling risk-aware design. Finally, we illustrate the framework on unmanned aerial vehicle (UAV) co-design with uncertain task profiles and specifications, revealing trade-offs that intervals cannot capture.

## C. Structure of the paper

Sections II and III review mathematical preliminaries, monotone co-design, and interval uncertainty. They are existing materials, except the formalization of **implementations** (design choices), adapted from the literature to better resemble the new constructions. Section IV develops the distributional and adaptive extension (distributions over DPs, implementations/specifications, queries/observations) using Quasi Universal Spaces; within it we review Quasi Measurable Spaces/Quasi Universal Spaces from [37], [38], while everything from Definition 21 onwards is new. Section V presents the UAV case study. Every imported definition and result carries a citation on its statement.

## II. MATHEMATICAL PRELIMINARIES

Some results in co-design and Quasi Universal Spaces are related to *category theory*, but it is *not* prerequisite here: every definition, statement, and proof in the main text uses ordinary set- and measure-theoretic language, and categorical observations appear only in remarks marked (CT), skippable without loss of continuity. Standard references are [39]–[41], and [3] for an applied, engineering-oriented introduction; This extended version collects the categorical concepts in full in Section VII, and extends [42].

1) *Sets and functions*:  $A \times B$  denotes the *Cartesian product*, with elements the tuples  $\langle m_A, m_B \rangle$ ,  $m_A \in A$ ,  $m_B \in B$ . Nested and unwrapped products are identified:  $A \times (B \times C)$  and  $A \times B \times C$ , and element-wise  $\langle m_A, \langle m_B, m_C \rangle \rangle$  and  $\langle m_A, m_B, m_C \rangle$ .

We write  $f: A \rightarrow B$  for a function (or *map*) with *domain*  $A$  and *co-domain*  $B$ , and  $m_A \mapsto f(m_A)$  for its action on elements. We denote the graph of  $f$  by  $\Gamma_f \stackrel{\text{def}}{=} \{\langle m_A, f(m_A) \rangle\} \subseteq A \times B$ , the image of  $X_A \subseteq A$  by  $f(X_A) \stackrel{\text{def}}{=} \{m_B \in B \mid \exists m_A \in X_A, f(m_A) = m_B\}$ , the pre-image of  $Y_B \subseteq B$  by  $f^{-1}(Y_B) \stackrel{\text{def}}{=} \{m_A \in A \mid f(m_A) \in Y_B\}$ , and the identity map by  $\text{id}_A: m_A \mapsto m_A$ . The *composite* of  $f: A \rightarrow B$  and  $g: B \rightarrow C$  is  $g \circ f: m_A \mapsto g(f(m_A))$ , drawn as  $A \xrightarrow{f} B \xrightarrow{g} C$ . A *partial map*  $f: A \rightarrow B$  is a map  $A|_f \rightarrow B$  defined on a subset  $A|_f \subseteq A$ . *Projections* and *permutations* are indicated by suffixes, e.g.  $\text{proj}_{1,3}: \langle m_A, m_B, m_C \rangle \mapsto \langle m_A, m_C \rangle$  and  $\text{perm}_{1 \leftrightarrow 2}: \langle m_A, m_B, m_C \rangle \mapsto \langle m_B, m_A, m_C \rangle$ .

We distinguish the *product*  $f \times g: \langle m_A, m'_A \rangle \mapsto \langle f(m_A), g(m'_A) \rangle$  from the *tuple*  $\langle f, g \rangle: m_A \mapsto \langle f(m_A), g(m_A) \rangle$  of maps; both can be indexed,  $\times_{i \in I} A_i$  and  $\langle f_i \rangle_{i \in I}$ .  $h(-, m_B): m_A \mapsto h(m_A, m_B)$  is the *curry* of  $h: A \times B \rightarrow C$ .

2) *Background on orders*: Orders on resources, functionalities, designs, etc., play a key role in co-design theory.

**Definition 1** (Poset and monotone maps). A *partially ordered set* (poset) is a tuple  $\mathcal{P} = \langle P, \preceq_P \rangle$  with  $P$  a set and  $\preceq_P$

a partial order (reflexive, transitive, antisymmetric relation); when clear we write  $P$  and  $\preceq$  as the poset and order. A map  $f: P \rightarrow Q$  is *monotone* if  $x \preceq_P y \Rightarrow f(x) \preceq_Q f(y)$ .

The *opposite* of a poset  $\mathcal{P} = \langle P, \preceq_P \rangle$  is the poset  $\mathcal{P}^{\text{op}} \stackrel{\text{def}}{=} \langle P, \preceq_P^{\text{op}} \rangle$  with the same elements and reversed ordering:  $x_P \preceq_P^{\text{op}} y_P \Leftrightarrow y_P \preceq_P x_P$ . The *product* of  $\langle P, \preceq_P \rangle$  and  $\langle Q, \preceq_Q \rangle$  is  $\langle P \times Q, \preceq_{P \times Q} \rangle$ , with

$$\langle x_P, x_Q \rangle \preceq_{P \times Q} \langle y_P, y_Q \rangle \Leftrightarrow (x_P \preceq_P y_P) \wedge (x_Q \preceq_Q y_Q).$$

Monotonicity is preserved by composition and products [3], and intervals are  $[x_{P,L}, x_{P,U}] \stackrel{\text{def}}{=} \{x_P \mid x_{P,L} \preceq x_P \preceq x_{P,U}\}$  for  $x_{P,L} \preceq x_{P,U}$ .

**Definition 2** (Upper closure and upper sets). The *upper closure* of  $X_P \subseteq P$  is  $\uparrow X_P \stackrel{\text{def}}{=} \{x_P \in P \mid \exists y_P \in X_P : y_P \preceq_P x_P\}$ , and  $X_P$  is an *upper set* if  $\uparrow X_P = X_P$ .  $\mathcal{U}(P)$  denotes its upper sets; *lower closures* and *lower sets* are dual.

**3) Measure theory and probability:** We recall standard measure theory and probability [43]. Additional structures (sigma algebras, topologies) are omitted when clear from context.

**Definition 3** (Measurable spaces and maps). A *sigma algebra*  $\Sigma_A$  on a set  $A$  is a collection of subsets containing  $A$  and closed under complements and countable unions;  $\langle A, \Sigma_A \rangle$  is a *measurable space* and sets in  $\Sigma_A$  are *measurable (events)*. A map between measurable spaces is *measurable* if pre-images of measurable sets are measurable.

For any family of subsets  $\mathcal{G} \subseteq 2^A$ , the sigma algebra *generated* by  $\mathcal{G}$ , denoted  $\sigma(\mathcal{G})$ , is the smallest sigma algebra containing  $\mathcal{G}$ . Completion maximally enlarges the events on which distributions evaluate, by adding subsets of events with zero probability.

**Definition 4** (Completion of a sigma algebra). Given a distribution  $\mu$  on measurable space  $\langle A, \Sigma \rangle$ , the Lebesgue completion  $\Sigma_\mu$  is the smallest sigma algebra containing  $\Sigma$  with  $Y \in \Sigma_\mu$  whenever  $Y \subseteq X \in \Sigma$  and  $\mu(X) = 0$ . The universal completion is the intersection over all distributions on it:

$$(\Sigma)_G \stackrel{\text{def}}{=} \cap \{\Sigma_\mu \mid \mu \text{ is a Distribution on } \langle A, \Sigma \rangle\}.$$

*Markov kernels* come from categorical probability and support compositional, graphical languages [44].

**Definition 5** (Markov kernels and their composition). A *Markov kernel* between measurable spaces,  $a: A \Rightarrow B$ , is a map  $a: A \times \Sigma_B \rightarrow [0, 1]$  s.t.:

- 1)  $a(m_A, -)$  is a probability distribution on  $\langle B, \Sigma_B \rangle$  for each  $m_A \in A$ ;
  - 2)  $a(-, Y_B): A \rightarrow [0, 1]$  is measurable for each  $Y_B \in \Sigma_B$ .
- We write  $a(m_A)$  for the distribution  $a(m_A, -)$ . Kernels  $a: A \Rightarrow B$  and  $b: B \Rightarrow C$  *compose* into  $b \circ a: \langle m_A, Z_C \rangle \mapsto \int_{m_B \in B} b(m_B, Z_C) a(m_A, dm_B)$ , with diagrammatic notation  $A \xRightarrow{a} B \xRightarrow{b} C$  in top-down and left-to-right form (Fig. 1). A distribution  $\mu$  on  $B$  is a kernel  $\{\star\} \xRightarrow{\text{const}_\mu} B$ , with  $b \circ \mu$  the pushforward.

Graphs, images, and pre-images of measurable maps require topological and descriptive-set structure.

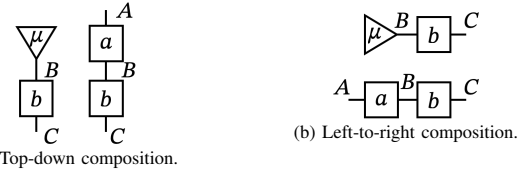


Fig. 1: Markov kernel diagrams  $b \circ \mu$  and  $b \circ a$  in two orientations.

**Definition 6** (Topological spaces, continuous maps, and Borel sigma algebras). A topological space is a tuple  $\langle A, \mathcal{O} \rangle$  with  $\mathcal{O}$  the open sets of  $A$ , containing the empty set and closed under arbitrary unions and finite intersections; *neighborhoods* of  $m_A \in A$  are open sets containing it. A map between topological spaces is *continuous* if pre-images of open sets are open. The Borel sigma algebra  $\Sigma_{\mathcal{O}}$ , with its elements called Borel sets, is the smallest sigma algebra containing  $\mathcal{O}$ .

Bool is the two-element set of logic true ( $\top$ ) and false ( $\perp$ ) with the discrete topology. *Homeomorphisms* are continuous bijections whose domain and co-domain are homeomorphic; homeomorphic spaces are topologically equivalent. Sets in  $(\Sigma_{\mathcal{O}})_G$  are called *universally measurable*.

**Definition 7** (Hausdorff, Polish, Souslin spaces and analytic sets). A Hausdorff space is a topological space in which distinct points have disjoint neighborhoods, and a Polish space one homeomorphic to a closed subset of  $\mathbb{R}^{\mathbb{N}}$  (e.g.  $\mathbb{R}$ ,  $\mathbb{R}^n$ ). An analytic set is a continuous image of a Polish space into a Hausdorff space; a Souslin space is an analytic Hausdorff space.

Every Borel set is analytic and every analytic set is universally measurable; countable intersections and unions of analytic sets are analytic, though complements need not be, and every continuous map is Borel measurable [43]. The next two theorems restate [43, Lemma 6.7.1, Theorem 6.7.3].

**Theorem 1.** Let  $A, B$  be Souslin. The graph  $\Gamma_f$  of a Borel measurable  $f: A \rightarrow B$  is Borel, hence analytic; conversely, if  $\Gamma_f$  is analytic then  $f$  is Borel measurable.

**Theorem 2.** For Borel measurable  $f: A \rightarrow B$  between Souslin spaces, images and pre-images of analytic sets are analytic.

### III. BACKGROUND OF MONOTONE CO-DESIGN

#### A. Monotone co-design theory

Co-design compositionally models and analyzes complex engineering systems with heterogeneous components and tightly coupled design choices. Its basic objects are *functionalities* and *resources*, each a poset. Notably, *functionalities* and *resources* are not properties of a poset but *roles* it plays at a component's interface: a component provides its functionalities while requiring its resources, and a *functionality* provided by one component can be the *resource* required by another (or by itself), as captured by the series and trace operations in Definition 9. Order reflects quality: larger means better, i.e. harder to provide as a *functionality*, more desirable as a *resource*. These need not be quantities that the designer controls (money, computation, etc.); they capture anything about performance, such as the weather or the task profile described in Section V.



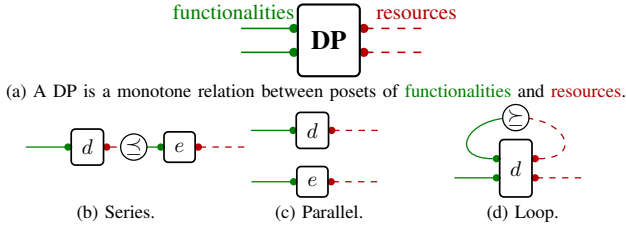


Fig. 2: DPs can be composed in different ways.

As a running example, consider UAV design, where perception is a key sub-system. Fixing all other subsystems, perception provides *detection accuracy* at the cost of *computation power* under given *weather conditions*. *Detection accuracy* can be a *resource* required by the planner, and *computation power* a *functionality* provided by the computer, forming two interconnections. Accuracy and computation power are positive reals, while weather conditions form a poset with non-comparable elements: a clear night and a foggy day pose qualitatively different challenges, and designs optimized for one need not perform well in the other. We then define design problems as descriptions of component performance relative to a given *functionality/resource* interface.

**Definition 8** (Design problem (DP)). Given posets  $F$  and  $R$  of *functionalities* and *resources*, a *DP* is an upper set of  $F^{\text{op}} \times R$ . We denote the set of such DPs by  $\text{DP}\{F, R\}$ . Given a DP denoted by  $\text{dp}$ , a pair  $\langle x_F, x_R \rangle$  of functionality  $x_F$  and resource  $x_R$  is *feasible* if  $\langle x_F, x_R \rangle \in \text{dp}$ . We order  $\text{DP}\{F, R\}$  by inclusion:  $\text{dp}_a \preceq \text{dp}_b \Leftrightarrow \text{dp}_a \subseteq \text{dp}_b$ .

The upper set condition captures a design intuition: if  $x_R$  suffices to provide  $x_F$  (i.e.,  $\langle x_F, x_R \rangle \in \text{dp}$ ), then a better resource  $x'_R \succeq x_R$  suffices for a worse functionality  $x'_F \preceq x_F$ . Taking the opposite order on  $F$  but not on  $R$  captures the difference between the two interfaces: for *this component* it is *easier* to provide *worse functionalities* with *better resources*. A DP therefore records *feasibility*, not *efficiency*: a design delivering *2 kg* of payload also appears as delivering *1 kg* at the same cost. Optimality is recovered by infimals of query results (Definition 12), not by inspection.

Systems performance is built from those of sub-systems as multi-graphs of DPs, called *co-design problems*; the composition operations are in Definition 9, some shown in Fig. 2.

**Definition 9** (Composition operations for DPs). The following operations construct new DPs from existing ones.

*Series*: for DPs  $\text{dp}_a \in \text{DP}\{P, Q\}$ ,  $\text{dp}_b \in \text{DP}\{Q, R\}$ , their series connection  $\text{dp}_a \circ \text{dp}_b$  is the DP

$$\{\langle x_P, x_R \rangle \mid \exists x_Q : \langle x_P, x_Q \rangle \in \text{dp}_a \text{ and } \langle x_Q, x_R \rangle \in \text{dp}_b\}.$$

Here  $\text{dp}_a$  uses the functionalities of  $\text{dp}_b$  as its resources.

*Parallel*: for  $\text{dp}_a \in \text{DP}\{P, Q\}$ ,  $\text{dp}'_a \in \text{DP}\{P', Q'\}$ , their parallel connection  $\text{dp}_a \otimes \text{dp}'_a$  is the DP

$$\{\langle \langle x_P, x'_P \rangle, \langle x_Q, x'_Q \rangle \rangle \mid \langle x_P, x_Q \rangle \in \text{dp}_a, \langle x'_P, x'_Q \rangle \in \text{dp}'_a\}.$$

This represents two non-interacting systems.

*Feedback/Trace/Loop*: for  $\text{dp} \in \text{DP}\{P \times R, Q \times R\}$ , its trace  $\text{Tr}(\text{dp})$  is the DP

$$\{\langle x_P, x_Q \rangle \mid \exists x_R : \langle \langle x_P, x_R \rangle, \langle x_Q, x_R \rangle \rangle \in \text{dp}\}.$$

Here  $\text{dp}$  uses its own functionalities as resources.

*Union/intersection*: for  $\text{dp}_a, \text{dp}_b \in \text{DP}\{P, Q\}$ , the union  $\text{dp}_a \vee$

$\text{dp}_b$  represents a free choice between them:

$$\{\langle x_P, x_Q \rangle \mid \langle x_P, x_Q \rangle \in \text{dp}_a \text{ or } \langle x_P, x_Q \rangle \in \text{dp}_b\};$$

while intersection  $\text{dp}_a \wedge \text{dp}_b$  has feasibilities promised by both:

$$\{\langle x_P, x_Q \rangle \mid \langle x_P, x_Q \rangle \in \text{dp}_a \text{ and } \langle x_P, x_Q \rangle \in \text{dp}_b\}.$$

Union and intersection extend to sets of DPs, e.g.  $\bigvee \{\text{dp}_i\}_{i \in I}$ .

**Remark 1.** (CT) With posets as *objects* (interfaces of composition) and DPs as *morphisms* (composable processes), the operations in Definition 9 make the resulting category *traced symmetric monoidal* (it has a “place side by side” product modeling parallel composition, in which the order of parallel components is immaterial and loops can be closed by a feedback operator). Moreover, for fixed  $F, R$  the hom-poset  $\text{DP}\{F, R\}$  (all morphisms between a fixed pair of objects, here carrying an order) is a *complete lattice* (every subset has a supremum and an infimum), and the category is enriched in  $\text{Pos}$  (its hom-sets themselves carry an order), the category of posets and monotone functions [2].

Designers care not only which functionality/resource pairs are achievable but which choices realize them, captured by *implementations*; we color *implementations* and later defined *specifications* similar to *functionalities* and *resources*. The definition below differs from [1], [2]: it better resembles the new parameterized framework and is a more effective deterministic baseline.

**Definition 10** (Monotone design problem with implementation (MDPI)<sup>1</sup>). For posets  $F, R$ , an MDPI is a tuple  $\langle I, \text{realize} \rangle$  with  $I$  a set of *implementations* (design choices) and  $\text{realize} : I \rightarrow \text{DP}\{F, R\}$ . We denote  $\text{realize}(i)$  by  $\text{dp}_i$ , and call

$$\text{dp}_{\langle I, \text{realize} \rangle} \stackrel{\text{def}}{=} \bigvee \{\text{dp}_i\}_{i \in I} \in \text{DP}\{F, R\}$$

the DP *induced* by  $\langle I, \text{realize} \rangle$ .

Here  $\text{realize}(i)$  is the trade-off provided by  $i$  and  $\text{dp}_{\langle I, \text{realize} \rangle}$  the *free choice* among implementations:  $\langle x_F, x_R \rangle \in \text{dp}_{\langle I, \text{realize} \rangle}$  iff  $\langle x_F, x_R \rangle \in \text{dp}_i$  for some  $i$ . For UAV perception, hardware and software choices such as *LiDAR models* and *object detection algorithms* each yield a different trade-off between *detection accuracy*, *computation power*, and *weather conditions*; every combination is an implementation  $i$  with trade-off  $\text{dp}_i$ . MDPIs compose as DPs do, a composite’s choice being a tuple of its components’; richer decision processes are a key contribution of this paper (Section IV-B).

**Definition 11.** The composition operations in Definition 9, lifted to MDPIs, are defined as follows. Each binary operation  $\diamond$  (one of  $\circ, \otimes, \vee$ , or  $\wedge$ ) is lifted to:

$$\begin{aligned} \langle I_1, \text{realize}_1 \rangle \diamond \langle I_2, \text{realize}_2 \rangle \\ \stackrel{\text{def}}{=} \langle I_1 \times I_2, i_1 \times i_2 \mapsto \text{realize}_1(i_1) \diamond \text{realize}_2(i_2) \rangle. \end{aligned}$$

The trace operation  $\text{Tr}$  is lifted to:

$$\text{Tr}(\langle I, \text{realize} \rangle) \stackrel{\text{def}}{=} \langle I, i \mapsto \text{Tr}(\text{realize}(i)) \rangle.$$

We reuse the notation of Definition 9 for the lifted operations.

<sup>1</sup>Note that DPs are monotone by definition. The term “monotone” here is inherited from monotone co-design [1].



These are well-defined: each output pairs a set with a map into  $\text{DP}\{F, R\}$ , since  $\text{realize}_1(i_1) \diamond \text{realize}_2(i_2)$  and  $\text{Tr}(\text{realize}(i))$  are DPs, so the output is again an MDPI. No order need be preserved, as implementation sets carry none; hence this case is simpler than the interval case below.

**Remark 2.** (CT) With the lifted operations of Definition 11 one might hope posets and MDPIs form a strict traced symmetric monoidal category. With ordinary set products, however, the coherence diagrams commute only up to canonical 2-isomorphisms (invertible transformations between morphisms), giving a bicategorical structure. The strictified set product of [3] yields a (strict) category, but the order of implementation sets in Cartesian products still prevents strict functoriality of  $\otimes$ .

This framework supports practical questions about optimal design choices, formalized as *queries*: they return all combinations of resources and design choices fulfilling the requested functionalities.

**Definition 12** (Queries for MDPIs). For an MDPI  $\langle I, \text{realize} \rangle$  with functionality and resource posets  $F, R$ , the *Fix function-alities minimize resources* query is

$$\text{FixFunMinRes}_{\langle I, \text{realize} \rangle} : A_F \mapsto \{ \langle x_R, i \rangle \mid \forall x_F \in A_F, \langle x_F, x_R \rangle \in \text{realize}(i) \}.$$

**Remark 3.** Designers care about *optimal* resources and the choices realizing them, suggesting restriction to the *minimal elements* of  $\text{proj}_1(\text{FixFunMinRes}_{\langle I, \text{realize} \rangle}(A_F))$ . A feasible resource set however need not be the upper closure of its minimal elements (e.g.  $(1, \infty)$  has none), so the framework computes all feasible pairs as in Definition 12. Real requirements are moreover often *functionality/resource pairs*, e.g. 90% accuracy on *sunny days* and 80% on *foggy days* while minimizing *computation power*; these are allowed by taking the *opposite* of a *resource* poset and treating it as a *functionality*.

### B. Interval uncertainty in co-design

In [2], [9], uncertainty is modeled via *intervals* of DPs, i.e., *optimistic* (best-case) and *pessimistic* (worst-case) designs; recall that  $\text{DP}\{F, R\}$  is ordered by inclusion, so a “better” (“larger”) DP makes more  $\langle x_F, x_R \rangle$  pairs feasible.

**Definition 13** (Interval uncertainty of DPs). Given posets  $F$  and  $R$ , the set of interval DPs  $\mathcal{I}(\text{DP}\{F, R\})$  is defined as

$$\{ [dp_L, dp_U] \mid dp_L, dp_U \in \text{DP}\{F, R\}, dp_L \preceq dp_U \},$$

where the lower bound  $dp_L$  represents the pessimistic estimate, and  $dp_U$  represents the optimistic estimate. Compositional operations in Definition 9 are lifted to intervals of DPs:

$$\begin{aligned} \text{Tr}_{\mathcal{I}}([dp_L, dp_U]) &\stackrel{\text{def}}{=} [\text{Tr}(dp_L), \text{Tr}(dp_U)], \\ [dp_L, dp_U] \diamond_{\mathcal{I}} [dp'_L, dp'_U] &\stackrel{\text{def}}{=} [dp_L \diamond dp'_L, dp_U \diamond dp'_U], \end{aligned}$$

where  $\diamond$  is either  $\circ$ ,  $\otimes$ ,  $\vee$  or  $\wedge$ . Moreover,  $\text{DP}\{F, R\}$  embeds into  $\mathcal{I}(\text{DP}\{F, R\})$  by  $dp \mapsto [dp, dp]$  [9].

For UAV sensing, a design combination might guarantee *detection accuracy* between 80% and 85%: the *lower bound*

$dp_L$  makes accuracies up to 80% feasible, the *upper bound*  $dp_U$  up to 85%.

Definition 13 carries a well-definedness obligation on which the whole interval framework rests: for  $\diamond_{\mathcal{I}}$  to land in  $\mathcal{I}(\text{DP}\{F, R\})$ , the pair produced must satisfy  $dp_L \preceq dp'_L$  and  $dp_U \preceq dp'_U$ . It does, because every operation in Definition 9 is *monotone* for the inclusion order, so  $dp_L \preceq dp'_L$  and  $dp'_L \preceq dp'_U$  give the inequality, and likewise for  $\text{Tr}$  [2], [9]. Monotonicity is thus what makes worst- and best-case bounds composable; we need it again in Lemma 8.

MDPIs lift to interval uncertainty too.

**Definition 14** (MDPIs with interval uncertainty). For posets  $F, R$ , an *MDPI with interval uncertainty* is a tuple  $\langle I, \text{realize} \rangle$  with  $I$  a set of *implementations* and  $\text{realize} : I \rightarrow \mathcal{I}(\text{DP}\{F, R\})$ . Compositions are formed by replacing each operation in Definition 11, e.g.  $\text{Tr}$ , with its interval lift from Definition 13, e.g.  $\mathcal{I}(\text{Tr})$ .

These are well-defined for both reasons combined: a composite’s implementation set is again a set, and the interval-valued map lands in  $\mathcal{I}(\text{DP}\{F, R\})$  by monotonicity. Queries then yield optimistic and pessimistic answers [9]; run Definition 12 on both bounds and read the resource sets as best- and worst-case designs.

## IV. DISTRIBUTIONAL UNCERTAINTY AND ADAPTIVE DECISIONS IN CO-DESIGN

Real design processes are more complex than one-shot *implementation* choices under best- and worst-case uncertainty: *implementations* carry parameters inducing *specifications* uncertain in ways intervals cannot capture, and some decisions can wait until part of that uncertainty resolves.

Consider again the perception and actuator sub-systems of a UAV. A sensor is chosen from a catalogue, each variant posing *specifications* such as field of view, noise characteristics, and failure modes. Manufacturers guarantee only *distributions* over these parameters, and purchasing a sensor draws a sample from them. Actuator *specifications* are likewise observable only after obtaining a concrete product, and the designer is often free to select one component *after* observing another, for instance buying the actuator after testing the delivered sensor. The design process is therefore inherently *adaptive* and involves full probability distributions, not only interval bounds; reducing them to confidence intervals would discard exactly the quantitative trade-offs between *functionality*, *resources*, and risk that practitioners care about.

To support such trade-offs compositionally we move from intervals to *distributions over DPs*, and from static MDPIs to MDPIs with *distributional uncertainty and adaptive decisions*. As in stochastic programming we take the distributions as given, agnostic to what they represent; in system design they arise from at least four sources:

- (i) *Manufacturing variability*: unit-to-unit spread of nominally identical components (motor constants, battery capacity), often a datasheet tolerance.
- (ii) *Environmental variability*: conditions outside control of the designer such as weather, temperature, or the *task*

*profile* of Section V, typically estimated from logged operational data.

- (iii) *Algorithmic randomness*: stochastic performance of sampling- or learning-based components, characterized by benchmarking over a test distribution.
- (iv) *Epistemic uncertainty*: limited knowledge of a component, e.g. a surrogate fitted to few measurements, the distribution being a posteriori knowledge.

The first three are *aleatoric*, the last *epistemic*; our constructions do not distinguish them, since both yield a distribution over DPs and the composition results apply verbatim. What *does* matter is independence: Lemma 8 assumes the component sample spaces factor, true for independent physical origins (motor and battery tolerances) but not under a common cause (a shared ambient temperature), in which case the shared quantity should be the common parameter fed to both components. Estimating these distributions, updating by conditioning, and exploration–exploitation are future work.

### A. Distributions over design problems as uncertain design results

In deterministic co-design the design result for fixed *implementations* is a DP encoding *functionality/resource* trade-offs (Definition 8). To model distributional uncertainty we need *random* DPs, i.e. *distributions over DPs*, that are:

- (i) *Compositional*: the series, parallel, and feedback operations in Definition 9 remain measurable;
- (ii) *Semantically expressive*: each evaluation  $\text{ev}_{x_F, x_R} : \text{DP}\{F, R\} \rightarrow \text{Bool}$ ,  $\text{dp} \mapsto (\langle x_F, x_R \rangle \in \text{dp})$ , is measurable, so we can ask the probability that a *functionality/resource* pair (or countable family) is feasible;
- (iii) *Stochasticity of functionalities/resources*: the constant injection  $\text{cInj} : F \times R \rightarrow \text{DP}\{F, R\}$ ,  $\langle x_F, x_R \rangle \mapsto \downarrow\{x_F\} \times \uparrow\{x_R\}$ , is measurable, turning distributions over  $F \times R$  into distributions over DPs;
- (iv) *Compatible with co-design*: deterministic DPs embed as degenerate (delta) distributions.

A naive approach puts a sigma algebra on  $\text{DP}\{F, R\}$  and takes ordinary probability measures; Counterexample 1 shows the first three requirements already fail, whereas this section's construction satisfies all four.

**Counterexample 1** (Impossibility of sigma algebras on DPs). Equip  $\mathbb{R}$  with the discrete partial order,  $x \preceq y \iff x = y$ . Each measurable map  $f : \mathbb{R} \rightarrow \text{Bool}$  then defines a DP by:  $\text{dp}_f \stackrel{\text{def}}{=} \{\langle x, b \rangle \mid f(x) \preceq_{\text{Bool}} b\} \in \text{DP}\{\mathbb{R}, \text{Bool}\}$ . Write  $\text{Bool}^{\mathbb{R}}$  as the set of measurable maps  $\mathbb{R} \rightarrow \text{Bool}$ , and  $\varphi : \text{Bool}^{\mathbb{R}} \rightarrow \text{DP}\{\mathbb{R}, \text{Bool}\}$ ,  $f \mapsto \text{dp}_f$ . Suppose for contradiction a sigma algebra  $\sigma(\text{DP}\{\mathbb{R}, \text{Bool}\})$  makes (i) all operations of Definition 9, (ii) the evaluations  $\text{ev}_{x_F, x_R}$ , and (iii) the constant injection  $\text{cInj}$  measurable. Equip  $\text{Bool}^{\mathbb{R}}$  with the pullback sigma algebra  $\sigma(\text{Bool}^{\mathbb{R}}) \stackrel{\text{def}}{=} \{A \subseteq \text{Bool}^{\mathbb{R}} \mid \varphi(A) \in \sigma(\text{DP}\{\mathbb{R}, \text{Bool}\})\}$ . The composite of three measurable maps

$$(\text{Bool} \times \mathbb{R}) \times \text{Bool}^{\mathbb{R}} \xrightarrow{\text{cInj} \times \varphi} \text{DP}\{\text{Bool}, \mathbb{R}\} \times \text{DP}\{\mathbb{R}, \text{Bool}\} \xrightarrow{\text{ev}_{\perp, \perp}} \text{DP}\{\text{Bool}, \text{Bool}\} \xrightarrow{\text{ev}_{\perp, \perp}} \text{Bool}$$

is measurable. Fixing the first argument to  $\perp$ , the induced map sends  $\langle \langle \perp, x \rangle, f \rangle$  to  $\{\langle \perp, x \rangle\} \times \{f(x') \preceq_{\text{Bool}} b\}$ , then to  $\{\langle \perp, b \rangle \mid f(x) \preceq_{\text{Bool}} b\}$ , then to  $\neg f(x)$ , whose measurability is equivalent to that of the usual evaluation map  $\text{Bool}^{\mathbb{R}} \times \mathbb{R} \rightarrow \text{Bool}$ ,  $\langle f, x \rangle \mapsto f(x)$ . No sigma algebra on  $\text{Bool}^{\mathbb{R}}$  makes this evaluation measurable [45], forming a contradiction.

*Remark 4* (Why a new formalism is needed). It is worth pausing on why the distributional case demands machinery the interval case does not, since the two are often expected to be siblings. In dynamical systems a disturbance and a signal are objects of the same kind, both time-indexed trajectories, so robust and stochastic formulations share one language and differ only in how the disturbance is quantified. Here the object carrying the uncertainty is the DP itself: a *relation* between posets, i.e., a higher-order object. Intervals cope with this because DPs are ordered and the operations of Definition 9 are monotone, so bounds compose (Definition 13). Distributions do not: counterexample 1 shows that *no* sigma algebra on  $\text{DP}\{F, R\}$  supports even the three minimal requirements (i)–(iii), so the obstruction is a property of ordinary measure theory rather than an artifact of a particular construction. What is required is a setting in which map spaces are first-class objects and evaluation is itself a morphism. This is the same obstruction that blocks higher-order probability on function spaces, and Quasi Measurable Spaces/Quasi Universal Spaces are exactly such a setting [37], [38]: the framework below is not a generalization for its own sake, but close to the least structure under which “a distribution over a design problem” is definable at all while the co-design operations survive. The dividend is that everything downstream is elementary: adaptivity is Markov kernels (Definition 25), queries are maps out of the distribution space (Definition 26), and the designer writes components and a diagram as in the deterministic case, with the sources of randomness made explicit.

We review Quasi Measurable Spaces/Quasi Universal Spaces [37], [38] before constructing distributions on DPs.

#### 1) Quasi Measurable Spaces and Quasi Universal Spaces:

Probability theory fixes a sample space  $\Omega$  with a sigma algebra  $\Sigma_{\Omega}$  of events, then considers *random variables* from it into other *measurable spaces*. Quasi Measurable Space invert this: random variables are first-class and sigma algebras derived from them, resolving the impossibility above. Introduced as a categorical construction for *higher-order probability* over maps [37], many of their results are proved categorically; category theory stays non-prerequisite here, as we restate every result we need in set- and measure-theoretic language. Fix  $\langle \Omega, \Sigma_{\Omega} \rangle$  and write  $\Omega^{\Omega}$  for the measurable maps  $\Omega \rightarrow \Omega$ .

**Definition 15** (Quasi-measurable spaces, quasi-measurable maps, and induced sigma algebras). A *Quasi Measurable Space (QMS)* is a pair  $\langle A, A^{\Omega} \rangle$  with  $A$  a set and  $A^{\Omega}$  a distinguished set of *random variables*  $\Omega \rightarrow A$  such that:

- 1) constant maps  $\text{const}_{m_A} : \omega \mapsto m_A$  for all  $m_A \in A$ , lie in  $A^{\Omega}$ ;
  - 2)  $\alpha \circ \varphi \in A^{\Omega}$  for every  $\alpha \in A^{\Omega}$  and measurable  $\varphi \in \Omega^{\Omega}$ .
- When the random variables are clear we write “QMS  $A$ ”; the sample space with all measurable maps is itself a QMS. A map

$f: A \rightarrow B$  of QMSs is *quasi-measurable* if  $f \circ \alpha \in B^\Omega$  for all  $\alpha \in A^\Omega$ ;  $B^A$  denotes these. Every QMS induces a sigma algebra  $\mathcal{B}(A) \stackrel{\text{def}}{=} \{X_A \subseteq A \mid \forall \alpha \in A^\Omega, \alpha^{-1}(X_A) \in \Sigma_\Omega\}$ .

When  $A$  carries a natural sigma algebra, e.g. the Borel one of a Polish space, taking all measurable maps as random variables reconnects to ordinary probability: if  $\Omega$  is also Polish, the induced sigma algebra is the completion of the Borel one. Taking *all* maps  $\Omega \rightarrow A$  as random variables would leave very few maps *from*  $A$  quasi-measurable, just as the power set as sigma algebra leaves very few maps *into*  $A$  measurable.

**Definition 16** (Product of QMSs). The *product* of QMSs  $\langle A, A^\Omega \rangle, \langle B, B^\Omega \rangle$  is  $\langle A \times B, (A \times B)^\Omega \rangle$ , a random variable on  $A \times B$  being a tuple of one per factor:  $(A \times B)^\Omega \stackrel{\text{def}}{=} \{\langle \alpha_a, \alpha_b \rangle \mid \alpha_a \in A^\Omega, \alpha_b \in B^\Omega\}$ .

Function spaces now carry natural sets of random variables.

**Definition 17** (QMS of quasi-measurable maps and evaluation). As random variables for  $B^A$  we take those maps  $\beta: \Omega \rightarrow B^A$  for which  $\omega \mapsto \beta(\varphi(\omega))(\alpha(\omega))$  is a random variable in  $B^\Omega$  for every  $\alpha \in A^\Omega$  and  $\varphi \in \Omega^\Omega$ .

Crucially, evaluation  $\text{ev}: A \times B^A \rightarrow B$ ,  $\langle m_A, f \rangle \mapsto f(m_A)$ , is then quasi-measurable [38].

*Remark 5.* (CT) With the terminal QMS (a singleton), products, and the QMS of quasi-measurable maps, QMSs and quasi-measurable maps form a *Cartesian closed category* [38]; the next lemma is a consequence.

**Lemma 1.** The curry (partial evaluation)  $\bar{f}: A \rightarrow C^B$ ,  $m_A \mapsto f(m_A, -)$ , of a quasi-measurable  $f: A \times B \rightarrow C$  is quasi-measurable, as are the pre- and post-composition maps  $(-) \circ f: g \mapsto g \circ f$  and  $f \circ (-): h \mapsto f \circ h$ , of types  $C^B \rightarrow C^A$  and  $A^C \rightarrow B^C$ , for quasi-measurable  $f: A \rightarrow B$ .

To connect QMSs with standard probability and Markov kernels compositionally, we restrict to “good” sample spaces.

**Definition 18** (Quasi-universal spaces). A QMS  $\langle A, A^\Omega \rangle$  is a *Quasi Universal Space (QUS)* if the sample space  $\Omega$  is Polish, typically  $\Omega = \mathbb{R}^n$  with the usual topology, and completion of its Borel sigma-algebra.

This is not computationally limiting: randomness that is computable or arises from countably many experiments originates from such sample spaces, and all QMS constructions restrict straightforwardly to QUSs. We adopt push-forward of distributions over QUS [38].

**Definition 19** (Distributions over QUSs). Let  $\langle A, A^\Omega \rangle$  be a QUS. A *distribution over  $A$*  is an equivalence class  $\{\langle \mu, \alpha \rangle\} / \simeq$  with  $\mu$  a probability distribution on  $\Omega$  and  $\alpha \in A^\Omega$  a random variable. Two pairs are equivalent if  $\mu(\alpha^{-1}(X_A)) = \mu'(\alpha'^{-1}(X_A))$  for all  $X_A \in \mathcal{B}(A)$ , i.e. if they induce the same measure on the induced sigma algebra. We write  $\mathcal{P}(A)$  for the set of classes and  $\langle \mu, \alpha \rangle$  for a class when unambiguous. To make  $\mathcal{P}(A)$  itself a QUS, we take as random variables precisely the maps  $\alpha_* \kappa: \Omega \rightarrow \mathcal{P}(A)$ ,  $\omega \mapsto \langle \kappa(\omega), \alpha \rangle$ , where  $\alpha \in A^\Omega$  and  $\kappa \in \mathcal{P}(\Omega)^\Omega$  is a classical Markov kernel assigning each  $\omega$  a distribution  $\kappa(\omega)$  on  $\Omega$ . One

can check that  $\mathcal{P}(\Omega)$  (with the induced sigma algebra  $\mathcal{B}(\Omega)$ ) is the usual space of probability measures on  $\Omega$  [38].

As in usual probability, (quasi) measurable deterministic maps induce maps between spaces of distributions.

**Lemma 2.** Each quasi-measurable  $f: A \rightarrow B$  induces  $\mathcal{P}(f): \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ ,  $\langle \mu, \alpha \rangle \mapsto \langle \mu, f \circ \alpha \rangle$ , transforming distributions by sampling an outcome and applying the map;  $\mathcal{P}(f)$  is quasi-measurable [38].

*Markov kernels* and their compositions carry over from classical measure theory.

**Definition 20** (Markov kernels for QUSs and their compositions). For QUSs  $A, B$ , a *Markov kernel*  $A \Rightarrow B$  is a quasi-measurable map  $a: A \rightarrow \mathcal{P}(B)$ . The *Kleisli composition*  $a \circ_P b: A \Rightarrow C$  of  $a: A \Rightarrow B$  and  $b: B \Rightarrow C$  is defined as follows. For  $m_A \in A$  write  $a(m_A) = \langle \mu, \alpha \rangle$ ; as  $b$  is quasi-measurable and  $\alpha \in B^\Omega$ , the composite  $\alpha \circ b$  lies in  $\mathcal{P}(C)^\Omega$ , so there are  $\kappa \in \mathcal{P}(\Omega)^\Omega$ ,  $\beta \in C^\Omega$  with  $\alpha \circ b = \beta_* \kappa$ . We set  $(a \circ_P b)(m_A) \stackrel{\text{def}}{=} \langle \mu \circ \kappa, \beta \rangle \in \mathcal{P}(C)$ .

**2) Sample space and assumptions on posets:** We fix a sample space  $\Omega$  (a QUS) aggregating all sources of foregoing randomness (Section IV-A) which is Polish; random variables into  $A$  are maps  $\Omega \rightarrow A$  in the QMS/QUS sense (not classical probability theory). We ask *which random variables into  $\text{DP}\{\mathcal{F}, \mathcal{R}\}$  are compositional, computable, and rich enough for applications*, under mild regularity on the posets involved.

**Assumption 1.** All posets considered are *Souslin spaces* and the order relation  $\{\langle x_P, x'_P \rangle \mid x_P \preceq_P x'_P\} \subseteq P \times P$  is a Borel set.

It holds for finite posets with discrete topology,  $\mathbb{R}^n$  with the usual topology and product or discrete order, and complete separable metric spaces with closed order relations.

**Lemma 3.** If posets  $P, Q$  satisfy Assumption 1, so do  $P^{\text{op}}$  and  $P \times Q$ .

*Proof.* We show the sets are Souslin and the order relations Borel. Since  $\{\langle x_P, x'_P \rangle \mid x_P \succeq x'_P\} = \text{perm}_{1 \leftrightarrow 2}(\{\langle x'_P, x_P \rangle \mid x'_P \preceq x_P\})$  and permutations are homeomorphisms, Theorem 2 applies and the opposite order set  $\{\langle x_P, x'_P \rangle \mid x_P \succeq x'_P\}$  is a Borel set. For the product poset, product of two Souslin Spaces is Souslin, so we only need to prove that the set  $\{\langle x_P, x_Q, x'_P, x'_Q \rangle \mid x_P \preceq x'_P, x_Q \preceq x'_Q\}$  is a Borel set. Observe that

$$\begin{aligned} & \{\langle x_P, x_Q, x'_P, x'_Q \rangle \mid x_P \preceq x'_P, x_Q \preceq x'_Q\} \\ &= \text{perm}_{2 \leftrightarrow 3}(\{\langle x_P, x'_P, x_Q, x'_Q \rangle \mid x_P \preceq x'_P, x_Q \preceq x'_Q\}) \\ &= \text{perm}_{2 \leftrightarrow 3}(\{x_P \preceq x'_P\} \times \{x_Q \preceq x'_Q\}). \end{aligned}$$

Products of Borel sets are Borel and permutations are homeomorphisms, so the statements follow.  $\square$

**Lemma 4.** Let  $P$  satisfy Assumption 1. If  $A \subseteq P$  is analytic, so are its closures  $\uparrow A$  and  $\downarrow A$ .

*Proof.* Let  $O = \{\langle x_P, x'_P \rangle \mid x_P \preceq x'_P\} \subseteq P \times P$ . Then  $\uparrow A = \text{proj}_2(\text{proj}_1^{-1}(A) \cap O)$  is analytic by Theorem 2, continuity of projections, and closure of analytic sets under



finite intersection; the same argument on the opposite order gives  $\downarrow A$ .  $\square$

**3) Quasi-universal space of design problems:** We first define QUSs of DPs by choosing a family of random variables, the analogue of a sigma algebra, so distributions make sense and Lemma 2 applies.

**Definition 21** (QUS of DPs). Let posets  $F$  and  $R$  satisfy Assumption 1. The QUS of DPs from  $F$  to  $R$  is  $\langle \text{DP}_A\{F, R\}, \text{DP}_A\{F, R\}^\Omega \rangle$ , where (i)  $\text{DP}_A\{F, R\}$  is the set of DPs  $\text{dp} \in \text{DP}\{F, R\}$  expressible as an upper closure  $\text{dp} = \uparrow A$  of an analytic  $A \subseteq F^{\text{op}} \times R$ ; (ii) a random variable  $\alpha: \Omega \rightarrow \text{DP}_A\{F, R\}$  lies in  $\text{DP}_A\{F, R\}^\Omega$  if its *outcome feasible set* (OFS)

$$F_\alpha \stackrel{\text{def}}{=} \{ \langle \omega, x_F, x_R \rangle \mid \langle x_F, x_R \rangle \in \alpha(\omega) \} \subseteq \Omega \times F \times R$$

is analytic.

**Lemma 5.**  $\langle \text{DP}_A\{F, R\}, \text{DP}_A\{F, R\}^\Omega \rangle$  is a QUS.

*Proof.* As  $\Omega$  is Polish, it suffices to check the two QMS axioms: (i) constant maps are random variables; (ii) random variables are closed under precomposition with measurable maps on  $\Omega$ . For (i), let  $\text{dp} = \uparrow A$  with  $A$  analytic in  $F^{\text{op}} \times R$ . By Lemma 4,  $\text{dp}$  is analytic, hence  $F_\alpha = \{ \langle \omega, x_F, x_R \rangle \mid \langle x_F, x_R \rangle \in \text{dp} \} = \Omega \times \text{dp}$  is analytic. Thus the constant map  $\alpha(\omega) = \text{dp}$  lies in  $\text{DP}_A\{F, R\}^\Omega$ . For (ii), let  $\alpha \in \text{DP}_A\{F, R\}^\Omega$  and  $\varphi \in \Omega^\Omega$  be measurable. Define the measurable map  $\tilde{\varphi}: \Omega \times F \times R \rightarrow \Omega \times F \times R$ ,  $\langle \omega, x_F, x_R \rangle \mapsto \langle \varphi(\omega), x_F, x_R \rangle$ . Then:

$$\begin{aligned} F_{\varphi \circ \alpha} &= \{ \langle \omega, x_F, x_R \rangle \mid \langle x_F, x_R \rangle \in \varphi \circ \alpha(\omega) \} \\ &= \{ \langle \omega, x_F, x_R \rangle \mid \langle x_F, x_R \rangle \in \alpha(\varphi(\omega)) \} \\ &= \{ \langle \omega, x_F, x_R \rangle \mid \langle \varphi(\omega), x_F, x_R \rangle \in F_\alpha \} = \tilde{\varphi}^{-1}(F_\alpha), \end{aligned}$$

which is analytic as the pre-image of an analytic set under a measurable map, according to Theorem 2.  $\square$

**Remark 6.** Stochasticity on *functionalities* such as *demands*, or *resources* such as *weather conditions*, is encoded by a DP component carrying only the corresponding interface; the Task DP of Fig. 8 is an example.

By construction,  $\langle \text{DP}_A\{F, R\}, \text{DP}_A\{F, R\}^\Omega \rangle$  contains all constant DPs via constant random variables. Definition 21 imposes the fewest conditions on posets and random variables: the former maximizes expressiveness, the latter unifies heterogeneous descriptions of components sharing an interface into one space. For an actuator's *rotary speed/torque* trade-off, one manufacturer may report torque constant and winding resistance, another a parameterized *torque-speed* curve; each reduces to an analytic set as in Definition 21, which is what makes decisions *among* them expressible (Section IV-B).

**4) Compositionality of the QUS of design problems:** All co-design operations on DPs turn out to be quasi-measurable on  $\text{DP}_A\{F, R\}$ , hence lift to distributions via Lemma 2.

**Theorem 3.** All composition operations of Definition 9 (series, parallel, feedback, union, intersection) are quasi-measurable as maps between the corresponding QUSs of DPs. Consequently, these operations lift to distributions over DPs in QUS. The lifted union  $\mathcal{P}(\vee)$  is the design result when one

may choose the outcome to implement after observing the randomness; the lifted intersection  $\mathcal{P}(\wedge)$  applies when one must be robust to it.

*Proof.* We show that post-composing any operation of Definition 9 with random variables whose OFSs are analytic yields a random variable with analytic OFS.

**Series:** Let  $\alpha_a \in \text{DP}_A\{P, Q\}^\Omega$  and  $\alpha_b \in \text{DP}_A\{Q, R\}^\Omega$  with feasible sets  $F_{\alpha_a}$  and  $F_{\alpha_b}$ . Post-composing with  $\circ$  gives

$$\circ \circ \langle \alpha_a, \alpha_b \rangle: \Omega \rightarrow \text{DP}_A\{P, R\}, \omega \mapsto \alpha_a(\omega) \circ \alpha_b(\omega)$$

with the OFS (with  $\star = \langle \omega, x_P, x_R \rangle$ )

$$\begin{aligned} &\{ \star \mid \langle x_P, x_R \rangle \in \alpha_a(\omega) \circ \alpha_b(\omega) \} \\ &= \{ \star \mid \exists x_Q \in Q, \langle x_P, x_Q \rangle \in \alpha_a(\omega), \langle x_Q, x_R \rangle \in \alpha_b(\omega) \} \\ &= \{ \star \mid \exists x_Q \in Q, \langle \omega, x_P, x_Q \rangle \in F_{\alpha_a}, \langle \omega, x_Q, x_R \rangle \in F_{\alpha_b} \} \\ &= \{ \star \mid \exists x_Q \in Q, \langle \omega, x_P, x_Q, x_R \rangle \in \text{proj}_{1,2,3}^{-1}(F_{\alpha_a}), \langle \omega, x_P, x_Q, x_R \rangle \in \text{proj}_{1,3,4}^{-1}(F_{\alpha_b}) \} \\ &= \text{proj}_{1,2,4}^{-1}(\text{proj}_{1,2,3}^{-1}(F_{\alpha_a}) \cap \text{proj}_{1,3,4}^{-1}(F_{\alpha_b})), \end{aligned}$$

which is analytic since Theorem 2 applies (projections are continuous) and set intersection preserves analytic.

**Parallel:** Let  $\alpha_a \in \text{DP}_A\{P, Q\}^\Omega$  and  $\alpha'_a \in \text{DP}_A\{P', Q'\}^\Omega$ , we similarly have the OFS of  $\otimes \circ \langle \alpha_a, \alpha'_a \rangle$  being analytic, with  $\star = \langle \omega, x_P, x'_P, x_Q, x'_Q \rangle$  and projection maps as above:

$$\begin{aligned} &\{ \star \mid \langle \langle x_P, x'_P \rangle, \langle x_Q, x'_Q \rangle \rangle \in \alpha_a(\omega) \otimes \alpha'_a(\omega) \} \\ &= \{ \star \mid \langle \omega, x_P, x_Q \rangle \in F_{\alpha_a}, \langle \omega, x'_P, x'_Q \rangle \in F_{\alpha'_a} \} \\ &= \text{proj}_{1,2,4}^{-1}(F_{\alpha_a}) \cap \text{proj}_{1,3,5}^{-1}(F_{\alpha'_a}). \end{aligned}$$

**Feedback/Trace/Loop:** Take  $\alpha \in \text{DP}_A\{P \times R, Q \times R\}^\Omega$  and the diagonal set  $D \stackrel{\text{def}}{=} \{ \langle \tilde{x}_R, \tilde{x}'_R \rangle \mid \tilde{x}_R = \tilde{x}'_R \}$ ,  $\text{Tr} \circ \alpha$  has OFS

$$\begin{aligned} &\{ \langle \omega, x_P, x_Q \rangle \mid \langle x_P, x_Q \rangle \in \text{Tr}(\alpha(\omega)) \} \\ &= \{ \langle \omega, x_P, x_Q \rangle \mid \exists x_R, \langle \omega, \langle x_P, x_R \rangle, \langle x_Q, x_R \rangle \rangle \in F_\alpha \} \\ &= \{ \langle \omega, x_P, x_Q \rangle \mid \exists x_R, x'_R, \langle \omega, x_P, x_R, x_Q, x'_R \rangle \in F_\alpha, \\ &\quad \langle \omega, x_P, x_R, x_Q, x'_R \rangle \in \{ \langle \tilde{\omega}, \tilde{x}_P, \tilde{x}_R, \tilde{x}_Q, \tilde{x}'_R \rangle \mid \tilde{x}_R = \tilde{x}'_R \} \} \\ &= \text{proj}_{1,2,4}^{-1}(F_\alpha \cap \text{proj}_{3,5}^{-1}(D)). \end{aligned}$$

It is an analytic set, since the diagonal set  $D \subseteq R \times R$  is a Borel set (thus analytic) and similar arguments as above apply.

**Union and Intersection:** For two random variables  $\alpha_a, \alpha_b \in \text{DP}_A\{P, Q\}^\Omega$ , post-composing union and intersection gives OFSs (with  $\star = \langle \omega, x_P, x_Q \rangle$ )

$$\begin{aligned} &\{ \star \mid \langle x_P, x_Q \rangle \in \alpha_a(\omega) \text{ or/and } \langle x_P, x_Q \rangle \in \alpha_b(\omega) \} \\ &= \{ \star \mid \langle \omega, x_P, x_Q \rangle \in F_{\alpha_a} \cup / \cap F_{\alpha_b} \} \\ &= F_{\alpha_a} \cup / \cap F_{\alpha_b}, \end{aligned}$$

both being analytic. Their semantics of choosing after observing the outcome, and being robust against two different outcomes, are self-explanatory in the expressions.  $\square$

**Remark 7.** (CT) With posets satisfying Assumption 1 as objects and QUSs of DPs as morphisms, Theorem 3 yields a symmetric monoidal category enriched in QUS. With additional structure from [38] it is even enriched in the Kleisli category of  $\mathcal{P}$ ; this abstraction is not needed here, but exemplifies [46], which systematically adds uncertainties to general categories.

**Remark 8.** Theorem 3 and Lemma 2 guarantee that the lifting can be performed, but are not constructive: they do not say how to compose two distributions, because of how product distributions are built in QUSs [38]. Composing  $\langle \mu_a, \alpha_a \rangle \in \mathcal{P}(\text{DP}_A\{P, Q\})$  in series with  $\langle \mu_b, \alpha_b \rangle \in \mathcal{P}(\text{DP}_A\{Q, R\})$  requires

$$\begin{aligned} & \mathcal{P}(\text{DP}_A\{P, Q\}) \times \mathcal{P}(\text{DP}_A\{Q, R\}) \\ & \xrightarrow{\otimes \mathcal{P}} \mathcal{P}(\text{DP}_A\{P, Q\} \times \text{DP}_A\{Q, R\}) \\ & \xrightarrow{\mathcal{P}(\S)} \mathcal{P}(\text{DP}_A\{P, R\}), \end{aligned}$$

i.e.,  $\langle \mu_a, \alpha_a \rangle \times \langle \mu_b, \alpha_b \rangle \mapsto \langle \mu, \alpha \rangle$  and then  $\langle \mu, \alpha \rangle \mapsto \langle \mu, \alpha \circ \S \rangle$ , where  $\langle \mu, \alpha \rangle$  induces on  $\text{DP}_A\{P, Q\} \times \text{DP}_A\{Q, R\}$  the product of the two factors' measures; it exists by [38, Theorem 5.29] and resembles the usual product distribution. If the sample space decomposes as  $\Omega = \Omega_a \times \Omega_b$  with  $\alpha_a(\langle \omega_a, \omega_b \rangle) = \alpha_{a,a}(\omega_a)$  and  $\alpha_b(\langle \omega_a, \omega_b \rangle) = \alpha_{b,b}(\omega_b)$ , however, then  $\langle \mu, \alpha \rangle = \langle \mu_{a,a} \times \mu_{b,b}, \alpha_{a,a} \times \alpha_{b,b} \rangle$  with  $\mu_{a,a}, \mu_{b,b}$  the marginals. As  $\Omega$  is usually a product of all randomness sources, such decompositions are common.

**5) Feasibility evaluation and constant injection:** With all **implementations** fixed, the design result is a distribution  $\langle \mu, \alpha \rangle \in \mathcal{P}(\text{DP}_A\{F, R\})$ , and one asks *what practical questions it answers*; the feasibility evaluation turns out to be quasi-measurable.

**Theorem 4.** The evaluation map

$$\text{ev}: \text{DP}_A\{F, R\} \times F^{\mathbb{N}} \times R^{\mathbb{N}} \rightarrow \text{Bool}$$

$\langle \text{dp}, \{x_{F,i}\}_{i \in \mathbb{N}}, \{x_{R,i}\}_{i \in \mathbb{N}} \rangle \mapsto (\forall i \in \mathbb{N}, \langle x_{F,i}, x_{R,i} \rangle \in \text{dp})$  is quasi-measurable.

*Proof.* Consider a random variable  $\alpha$  for  $\text{DP}_A\{F, R\} \times F^{\mathbb{N}} \times R^{\mathbb{N}}$ , with  $\alpha_1 \in \text{DP}_A\{F, R\}^{\Omega}$ ,  $\alpha_2 \in (F^{\mathbb{N}})^{\Omega}$ , and  $\alpha_3 \in (R^{\mathbb{N}})^{\Omega}$  as its components. Post-composing with  $\text{ev}$  gives

$$\text{ev} \circ \alpha: \Omega \rightarrow \text{Bool}, \omega \mapsto (\forall i, \langle \omega, \alpha_2(\omega)_i, \alpha_3(\omega)_i \rangle \in F_{\alpha_1}).$$

We need to prove this map lives in  $\text{Bool}^{\Omega}$ . Namely, the pre-image of  $\top \in \text{Bool}$ ,  $(\text{ev} \circ \alpha)^{-1}(\top)$ , is universally measurable. For each index  $i$ , we know the map  $\bar{\alpha}_{2,3,i}: \omega \mapsto \langle \omega, \alpha_2(\omega)_i, \alpha_3(\omega)_i \rangle$  is measurable, so the corresponding pre-image  $\bar{\alpha}_{2,3,i}^{-1}(F_{\alpha_1})$  is universally measurable.  $(\text{ev} \circ \alpha)^{-1}(\top) = \bigcap_{i \in \mathbb{N}} \bar{\alpha}_{2,3,i}^{-1}(F_{\alpha_1})$  is then universally measurable since it is a countable intersection of universally measurable sets.  $\square$

Theorem 4 thus lets us compute, from a distribution on DPs together with distributions over **functionalities** and **resources**, the probability that a countable family of functionality/resource pairs is feasible for a sampled DP; constant distributions recover evaluation at fixed levels.

Finally, we show that constant maps are quasi-measurable, so distributions on **functionalities/resources** forms distributions on DPs, further supporting Remark 6.

**Lemma 6.** The sample space  $\Omega$ , equipped with the discrete order  $\omega \preceq \omega' \iff \omega = \omega'$  satisfies Assumption 1. We always assign this discrete order to  $\Omega$  without further notice.

*Proof.* By definition of QUSs,  $\Omega$  is homeomorphic to  $\mathbb{R}$ , hence Souslin. The diagonal set  $\{\langle \omega, \omega \rangle\} \subseteq \Omega \times \Omega$  is closed, thus

analytic and Borel, and coincides with the order relation under the discrete order.  $\square$

**Theorem 5.** The constant injection map

$$\text{cInj}: F \times R \rightarrow \text{DP}_A\{F, R\}, \langle x_F, x_R \rangle \mapsto \downarrow \{x_F\} \times \uparrow \{x_R\}$$

is quasi-measurable.

*Proof.*  $\{\langle x_F, x_R \rangle\}$  is an analytic set in  $F^{\text{op}} \times R$ , so  $\text{cInj}$  is well-defined since  $\downarrow \{x_F\} \times \uparrow \{x_R\} = \uparrow \{\langle x_F, x_R \rangle\}$  in  $F^{\text{op}} \times R$  and Lemma 4 applies. Pick two random variables  $\alpha_a \in F^{\Omega}$  and  $\alpha_b \in R^{\Omega}$ . We can write the OFS  $F_{\text{cInj} \circ \langle \alpha_a, \alpha_b \rangle}$  as

$$\begin{aligned} & \{\langle \omega, x_F, x_R \rangle \mid x_F \preceq \alpha_a(\omega), \alpha_b(\omega) \preceq x_R\} \\ & = \{\langle \omega, x_F, x_R \rangle \mid \langle \omega, x_F \rangle \in \downarrow \Gamma_{\alpha_a}, \langle \omega, x_R \rangle \in \uparrow \Gamma_{\alpha_b}\} \\ & = \text{proj}_{1,2}^{-1}(\downarrow \Gamma_{\alpha_a}) \cap \text{proj}_{1,3}^{-1}(\uparrow \Gamma_{\alpha_b}) \end{aligned}$$

which is analytic by Theorems 1 and 2 and set intersection preserving analytic.  $\square$

**6) Confidence bounds on distributions over design problems:**

Distributions over DPs induce *confidence bounds*, generalizing interval uncertainty. As DPs are rarely totally ordered, we distinguish levels for sampling *inside* a bound from *outside*.

**Definition 22** (Inner confidence bounds). Let  $\langle \mu, \alpha \rangle \in \mathcal{P}(\text{DP}_A\{F, R\})$  and  $p \in [0, 1]$ . An interval  $[\text{dp}_L, \text{dp}_U]$  of DPs is an *inner confidence bound* of level  $p$  if some universally measurable  $X \subseteq \Omega$  satisfies  $X \subseteq \alpha^{-1}([\text{dp}_L, \text{dp}_U])$  and  $\mu(X) \geq p$ . If both the set inclusion and the inequality hold with equality, we call the bound *strict*.

**Definition 23** (Outer confidence bounds). Let  $\langle \mu, \alpha \rangle \in \mathcal{P}(\text{DP}_A\{F, R\})$  and  $p, q \in [0, 1]$ . An interval  $[\text{dp}_L, \text{dp}_U]$  of DPs is an *outer confidence bound* of level  $p, q$  if some universally measurable  $X, Y \subseteq \Omega$  satisfy  $X \supseteq \alpha^{-1}([\emptyset, \text{dp}_L])$  and  $\mu(X) \leq p$ ;  $Y \supseteq \alpha^{-1}([\text{dp}_U, F^{\text{op}} \times R])$  and  $\mu(Y) \leq q$ . If all inclusions and inequalities are equalities, the bound is *strict*.

Informally, an inner bound keeps the design result between the bounds with probability *at least*  $p$ , an outer bound keeps the probabilities of results worse or better than the boundaries *at most*  $p$  and  $q$ ; both are strict when exact.

**Remark 9.** Confidence bounds need not exist for a given  $\langle \mu, \alpha \rangle$  and, when they do, are generally not unique even among strict ones; under extra structure, e.g. ordered  $\Omega$  and monotone  $\alpha$ , they can often be built from subsets of  $\Omega$  such as intervals. Although intervals of DPs always compose by composing their bounds (Definition 13), the result need not be a valid confidence bound, since the lifted operations of Theorem 3 are non-constructive; when it is valid, monotonicity implies the level can only deteriorate. Inner bounds compose whenever the decomposition of Remark 8 holds; outer bounds are future work.

**Lemma 7.** For each map  $f: A \rightarrow B$  and subsets  $X_A \subseteq A$ ,  $Y_B \subseteq B$ , we have  $f(f^{-1}(Y_B)) \subseteq Y_B$  and  $X_A \subseteq f^{-1}(f(X_A))$ .

*Proof.* For each  $m_B$  in  $f(f^{-1}(Y_B))$ , we know there exist  $m_A$  in  $f^{-1}(Y_B)$  so that  $f(m_A) = m_B$ , directly giving  $m_B \in Y_B$  by the definition of  $f^{-1}(Y_B)$ . For each  $m_A$  in  $X_A$ , by the

definition of  $f(X_A)$  we have:  $f(m_A) \in f(X_A)$ . Then by the definition of  $f^{-1}(f(X_A))$ ,  $m_A \in f^{-1}(f(X_A))$ .  $\square$

**Lemma 8.** Suppose we have a diagram of DPs (for instance Fig. 4 in Section V), with  $f$  being the deterministic map taking the component DPs and returning the composed DP. Denote the distributions over DPs for each component as  $\langle \mu_i, \alpha_i \rangle$ , indexed by a finite set  $I$ . Suppose we have a decomposition of the sample space, namely:  $\Omega = \times_{i \in I} \Omega_i$ , and  $\alpha_i(\langle \omega_i \rangle_{i \in I}) = \alpha_{i,i}(\omega_i)$ . Given an inner confident bound  $[\text{dp}_{L,i}, \text{dp}_{U,i}]$  for each distribution, together with  $X_i \subseteq \Omega$  and  $p_i$  satisfying conditions in Definition 22, the interval composition

$$\mathcal{I}(f)(\langle [\text{dp}_{L,i}, \text{dp}_{U,i}] \rangle_{i \in I}) = [f(\langle \text{dp}_{L,i} \rangle), f(\langle \text{dp}_{U,i} \rangle)]$$

is an inner confidence bound of level  $\prod_{i \in I} p_i$  for the composed distribution.

*Proof.* The composed distribution reads:  $\langle \times_i \mu_{i,i}, f \circ \times_i \alpha_{i,i} \rangle$ . Let  $X_{i,i} \stackrel{\text{def}}{=} \text{proj}_i(X_i)$ . We will prove the set  $\times_i X_{i,i} \subseteq \Omega$  satisfies conditions in Definition 22. We have  $\text{proj}_i^{-1}(X_{i,i}) \supseteq X_i$  by Lemma 7. First observe:

$$\alpha_i^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}]) = \text{proj}_i^{-1}(\alpha_{i,i}^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}]))$$

and since  $X_i \subseteq \alpha_i^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}])$ ,

$$\begin{aligned} X_{i,i} = \text{proj}_i(X_i) &\subseteq \text{proj}_i(\alpha_i^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}])) \\ &= \text{proj}_i(\text{proj}_i^{-1}(\alpha_{i,i}^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}]))). \end{aligned}$$

With Lemma 7 we further have

$$\text{proj}_i(\text{proj}_i^{-1}(\alpha_{i,i}^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}]))) \subseteq \alpha_{i,i}^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}]),$$

thus  $X_{i,i} \subseteq \alpha_{i,i}^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}])$ .

Since all the compositional operations are monotone, we have

$$f(\times_i [\text{dp}_{L,i}, \text{dp}_{U,i}]) \subseteq [f(\langle \text{dp}_{L,i} \rangle), f(\langle \text{dp}_{U,i} \rangle)],$$

thus (first applying Lemma 7)

$$\begin{aligned} \times_i [\text{dp}_{L,i}, \text{dp}_{U,i}] &\subseteq f^{-1}(f(\times_i [\text{dp}_{L,i}, \text{dp}_{U,i}])) \\ &\subseteq f^{-1}([f(\langle \text{dp}_{L,i} \rangle), f(\langle \text{dp}_{U,i} \rangle)]), \end{aligned}$$

and then

$$\begin{aligned} \times_i X_{i,i} &\subseteq \times_i \alpha_{i,i}^{-1}([\text{dp}_{L,i}, \text{dp}_{U,i}]) \\ &= (\times_i \alpha_{i,i})^{-1}(\times_i [\text{dp}_{L,i}, \text{dp}_{U,i}]) \\ &\subseteq (\times_i \alpha_{i,i})^{-1}(f^{-1}([f(\langle \text{dp}_{L,i} \rangle), f(\langle \text{dp}_{U,i} \rangle)])) \\ &= (f \circ \times_i \alpha_{i,i})^{-1}([f(\langle \text{dp}_{L,i} \rangle), f(\langle \text{dp}_{U,i} \rangle)]). \end{aligned}$$

With

$$\mu_{i,i}(X_{i,i}) = \mu_i(\text{proj}_i^{-1}(X_{i,i})) \geq \mu_i(X_i) \geq p_i,$$

$$\times_i \mu_{i,i}(\times_i X_{i,i}) = \mu_{1,1}(X_{1,1}) \times \mu_{2,2}(X_{2,2}) \cdots \geq \prod_{i \in I} p_i,$$

$[f(\langle \text{dp}_{L,i} \rangle), f(\langle \text{dp}_{U,i} \rangle)]$  is an inner bound of level  $\prod_{i \in I} p_i$ .  $\square$

## B. Implementations, specifications, and adaptive decisions

In deterministic co-design MDPIs are design processes: choosing **implementations** yields deterministic DPs (Definition 10). Definition 21 lets us extend this to MDPIs with

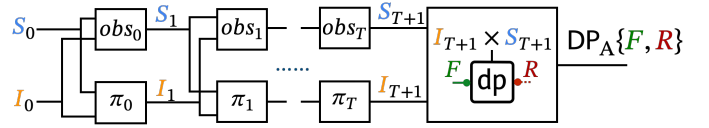


Fig. 3: Diagram of  $a \circ_{\mathcal{P}} \langle \text{obs}_T, \pi_T \rangle \circ_{\mathcal{P}} \cdots \circ_{\mathcal{P}} \langle \text{obs}_1, \pi_1 \rangle \circ_{\mathcal{P}} \langle \text{obs}_0, \pi_0 \rangle: S_0 \times I_0 \Rightarrow \text{DP}_A\{F, R\}$ , a  $T$ -stage adaptive decision process. The right-most kernel,  $a: S_{T+1} \times I_{T+1} \Rightarrow \text{DP}_A\{F, R\}$ , is illustrated in a similar way to those in Fig. 8.

*distributional uncertainty*, where design choices yield distributions on DPs and adaptivity is re-parameterization by Markov kernels in QUS (Definition 20).

**Definition 24** (MDPIs with distributional uncertainty). Let  $F, R$  satisfy Assumption 1. An MDPI with *distributional uncertainty* is a tuple  $\langle S, I, a \rangle$  with  $S, I$  QUSs of system **specifications** and **implementations** and  $a$  a Markov kernel  $a: S \times I \Rightarrow \text{DP}_A\{F, R\}$ , i.e. a quasi-measurable map  $S \times I \rightarrow \mathcal{P}(\text{DP}_A\{F, R\})$ .

For a one-shot (non-adaptive) UAV design,  $I$  holds all design decisions and  $S$  the specifications known *before* deciding:  $i$  is e.g. a combination of sensors, actuators, body, and algorithms,  $s$  e.g. sensing distances and mean/variance of actuator efficiency, and  $a(i, s)$  the induced distribution on design results. Although all uncertainty could be folded into  $\Omega$ , keeping **specifications** explicit is what lets us *re-parameterize* design processes and express multi-stage adaptive decisions.

**Definition 25** (Re-parameterization of distributional uncertain MDPIs). Given a distributional uncertain MDPI  $\langle S, I, a \rangle$ , re-parameterizing it with another Markov kernel  $f: S' \times I' \Rightarrow S \times I$  gives distributional uncertain MDPI  $\langle S', I', a \circ_{\mathcal{P}} f \rangle$ .

Re-parameterization captures *multi-stage adaptive decision processes*. At each stage  $k$ , specifications  $S_k$  and implementations  $I_k$  are determined from the previous stage via a Markov kernel  $\langle \text{obs}_{k-1}, \pi_{k-1} \rangle: S_{k-1} \times I_{k-1} \Rightarrow S_k \times I_k$  (Fig. 3). It has two components: (a) an *observation model*  $\text{obs}_{k-1}: S_{k-1} \times I_{k-1} \Rightarrow S_k$ , describing how the environment and specifications evolve (e.g. observing performance of the chosen actuator); (b) a *decision policy*  $\pi_{k-1}: S_{k-1} \times I_{k-1} \Rightarrow I_k$ , choosing the next implementation (possibly randomly) based on past specifications and decisions (e.g. choosing a sensing algorithm based on performance of the chosen sensor).

**Remark 10.** The space of Markov kernels  $\text{QUS}\{A, \mathcal{P}(B)\}$  is itself a QUS with quasi-measurable evaluation [38], so policies can be **implementations**; however parameterized policies are in practice more convenient.

**Remark 11.** The lifted operations of Theorem 3 and the re-parameterizations of Definition 25 give two layers of adaptivity: (1) the *policy level*, via the kernels  $\pi_k$ , and (2) the *co-design level*, since composition happens after all uncertainties relevant to the composite are observed. In interval co-design (Section III-B) the optimistic and pessimistic DPs are solved independently, so choices may differ between the two and are all made at the *co-design level*, after observing every relevant uncertainty.

Re-parameterizations can be represented by arbitrary string diagrams, as in Fig. 7.



**Remark 12.** (CT) With structural quasi-measurable maps (duplication, deletion),  $\mathcal{P}$  extends to a strong monad (adding structure compatible with series/parallel composition) on QMS [38], and restricting to QUSs makes it commutative, so its Kleisli category (morphisms carrying the added structure) is symmetric monoidal and arbitrary diagrams of Markov kernels are describable [37], [38]. Categorical coherence guarantees any diagram is a well-defined composite kernel [44], [47], as co-design diagrams like Fig. 4 (bottom) define a proper DP [2]; both structures organize into a 2-category with uncertain co-design problems as 1-cells and re-parameterizations as 2-cells [46].

Conditionals and disintegrations of Markov kernels are not part of our basic structure [38] but are powerful for learning and solving (hidden) Markov models [48]; with the literature on finite-horizon, varying-state-space decision processes [49]–[51] they are a natural basis for efficient algorithms, left as future work. Compositionality and re-parameterization thus give a quantitative, adaptive language: uncertain MDPIs are defined per component, then composed and re-parameterized by kernels representing the design process (Fig. 8).

### C. Queries and observations in co-design with distributional uncertainty

We now formalize what one can ask of distributional uncertain MDPIs: *queries* aggregate information about distributions over DPs, while *observations* assign a quantity to each individual DP and lift to queries via  $\mathcal{P}$ ; map spaces that QUSs provide greatly increase their flexibility (Example 2).

#### 1) Queries:

**Definition 26** (Queries for distributional uncertain MDPIs). For a distributional uncertain MDPI  $\langle S, I, a \rangle$  with posets  $F, R$ , a *query* is a map  $h: \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow M$  for some set  $M$ .

We do not require  $h$  quasi-measurable: querying is the final modeling step and measurability does not affect expressiveness. When analyzing or optimizing  $S \times I \xrightarrow{a} \mathcal{P}(\text{DP}_A\{F, R\}) \xrightarrow{h} M$  one may impose measurability, continuity, or convexity on  $h$  or  $h \circ a$ , depending on the task.

**Example 1** (Probability of feasible free choice). Currying the evaluation of Theorem 4 gives a query for the probability of providing a free choice among countably many **functionality/resource** pairs:  $\text{pr}: \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow ((F \times R)^{\mathbb{N}} \rightarrow [0, 1])$ ,  $\text{pr}(\langle \mu, \alpha \rangle)(\{ \langle x_{F,i}, x_{R,i} \rangle \}_{i \in \mathbb{N}}) = \mu(\{ \omega \mid \forall i, \langle x_{F,i}, x_{R,i} \rangle \in \alpha(\omega) \})$ .

**Example 2** (Fix functionalities, minimize resources (probabilistic)). The probabilistic *fix functionalities, minimize resources* query returns, for each countable family of required **functionalities**, the probability that each **resource** is sufficient:

$$\text{FixFunMinRes}: \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow (F^{\mathbb{N}} \rightarrow (R \rightarrow [0, 1])),$$

$$\langle \mu, \alpha \rangle \mapsto \left( \{ x_{F,i} \}_{i \in \mathbb{N}} \mapsto (x_R \mapsto \mu(\{ \omega \mid \forall i, \langle x_{F,i}, x_R \rangle \in \alpha(\omega) \})) \right).$$

Some questions are defined only for certain distributions, giving partial queries.

**Definition 27** (Partial queries). A *partial query* is a partial map  $h: \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow M$ .

**Example 3** (Confidence intervals and point distributions). Where confidence intervals (Definitions 22 and 23) can be computed, the map  $\text{conf}: \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow \mathcal{I}(\text{DP}\{F, R\})$  is a partial query; post-composing it with a deterministic map  $q: \text{DP}_A\{F, R\} \rightarrow M$  yields another partial query  $\mathcal{P}(\text{DP}_A\{F, R\}) \xrightarrow{\text{conf}} \mathcal{I}(\text{DP}\{F, R\}) \xrightarrow{\mathcal{I}(q)} \mathcal{I}(M)$ . A second is  $\text{const}_P^{-1}: \mathcal{P}(\text{const})(\mathcal{P}(F \times R)) \rightarrow \mathcal{P}(F \times R)$ , satisfying  $\text{const}_P^{-1} \circ \mathcal{P}(\text{const}) = \text{id}$  and sending each distribution on DPs induced by one on  $F \times R$  back to it.

**Remark 13.** A partial query  $h$  applied to a specific uncertain MDPI  $\langle S, I, a \rangle$  may still give a total map  $h \circ a: S \times I \rightarrow M$ , provided all design results lie in its domain,  $a(S \times I) \subseteq \mathcal{P}(\text{DP}_A\{F, R\})|_h$ .

#### 2) Observations:

**Definition 28** (Observations for realizations of MDPIs). For a distributional uncertain MDPI with posets  $F, R$ , an *observation* is a quasi-measurable  $r: \text{DP}_A\{F, R\} \rightarrow O$  into a QUS  $O$ .

**Remark 14.** Quasi-measurability enables learning and Bayesian updating (datasets, priors and posteriors, an aleatoric/epistemic split), and lifts every observation to a query  $\mathcal{P}(r): \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow \mathcal{P}(O)$  from distributions on DPs to distributions on outcomes. For a poset  $O$  and monotone  $r$ , an inner bound  $[\text{dp}_L, \text{dp}_U]$  induces  $[r(\text{dp}_L), r(\text{dp}_U)]$  at the same level.

Under mild conditions on  $R$  one can define an observation returning the *infimal resource requirement* for a set of functionalities.

**Lemma 9.** For every QMS  $P$ , the constant sequence injection  $\text{cInj}_{\mathbb{N}}: P \rightarrow P^{\mathbb{N}}, x_P \mapsto \{x_P\}_{i \in \mathbb{N}}$  is quasi-measurable by the definition of countable products of QMSs [38].

**Lemma 10.** Let  $P$  be a poset satisfying Assumption 1. If:

- 1) it is totally ordered.
- 2) each measurable set has an infimum. We denote  $-\infty \stackrel{\text{def}}{=} \inf(P)$  and  $\infty \stackrel{\text{def}}{=} \inf(\emptyset)$ .
- 3) the Borel sigma algebra is generated by open and unbounded open intervals, i.e.,  $(a, b)$ ,  $[-\infty, b)$ , and  $(a, \infty]$ .

Then the infimum map  $\inf: \mathcal{B}(P) \rightarrow P, A \mapsto \inf(A)$  is quasi-measurable, where the random variables  $\mathcal{B}(P)^{\Omega}$  are defined as in [38, Definition 2.47].

*Proof.* Let  $\beta$  be a random variable in  $\mathcal{B}(P)^{\Omega}$ . We need to prove  $\inf \circ \beta \in P^{\Omega}$ . Because of how the Borel sigma algebra is generated, we only need to prove pre-images of two types of intervals:  $[-\infty, b]$  and  $[\infty, b)$ , are universally measurable. From [38, Definition 2.47], we know that there exists  $D \in \mathcal{B}(\Omega \times P)$ , making  $x_P \in \beta(\omega) \iff \langle \omega, x_P \rangle \in D$ . We can compute the pre-image of  $[-\infty, b]$  by

$$\begin{aligned} & (\inf \circ \beta)^{-1}([-\infty, b]) \\ &= \{ \omega \mid \inf(\beta(\omega)) \preceq b \} \\ &= \{ \omega \mid \exists x_P \in \beta(\omega), x_P \preceq b \} \\ &= \{ \omega \mid \exists x_P, \langle \omega, x_P \rangle \in D, x_P \preceq b \} \end{aligned}$$

$$\begin{aligned}
&= \{\omega \mid \exists x_P, \langle \omega, x_P \rangle \in D, \langle \omega, x_P \rangle \in \text{proj}_2^{-1}([\infty, b])\} \\
&= \text{proj}_1(D \cap \text{proj}_2^{-1}([\infty, b])).
\end{aligned}$$

For the interval  $[\infty, b)$ , we only need to take the strict inequalities and have  $(\inf \circ \beta)^{-1}([-\infty, b)) = \text{proj}_1(D \cap \text{proj}_2^{-1}([\infty, b)))$ . Both of them are universally measurable by Theorem 2 and set intersection preserving analytic.  $\square$

Its conditions hold for any second countable totally ordered set with order-generated topology containing its limits, e.g.  $\mathbb{R}$  with  $\pm\infty$  or closed intervals of  $\mathbb{R}$ .

*Example 4* (Distribution of minimal resource). Suppose  $R$  satisfies the conditions of Lemma 10 and fix an at most countable family of functionalities  $\{x_{F,i}\}_{i \in \mathbb{N}}$ . We construct an observation  $r_{\min}: \text{DP}_A\{F, R\} \rightarrow R$  returning the infimal resource providing all of them. Currying  $\text{ev}$  of Theorem 4 gives a quasi-measurable  $\bar{\text{ev}}: \text{DP}_A\{F, R\} \rightarrow \text{Bool}^{F^{\mathbb{N}} \times R^{\mathbb{N}}}$  (Lemma 1); pre-composing with  $\text{const}_{\{x_{F,i}\}_{i \in \mathbb{N}}} \times \text{id}$  and then  $\text{const}_{\mathbb{N}}$  (Lemma 9) lands in  $\text{Bool}^R$ ; with the quasi-measurable bijection  $\chi$  of [38, Lemma 2.51] between  $\text{Bool}^R$  and  $\mathcal{B}(R)$  and the infimum map of Lemma 10, we obtain

$$\begin{aligned}
\text{DP}_A\{F, R\} &\xrightarrow{\bar{\text{ev}}} \text{Bool}^{F^{\mathbb{N}} \times R^{\mathbb{N}}} \\
&\xrightarrow{(-) \circ (\text{const}_{\{x_{F,i}\}_{i \in \mathbb{N}}} \times \text{id})} \text{Bool}^{R^{\mathbb{N}}} \\
&\xrightarrow{(-) \circ \text{cl}_{\text{Inf}_{\mathbb{N}}}} \text{Bool}^R \xrightarrow{\chi} \mathcal{B}(R) \xrightarrow{\inf} R.
\end{aligned}$$

On  $\text{dp} \in \text{DP}_A\{F, R\}$  this returns

$$r_{\min}(\text{dp}) = \inf \left( \{x_R \mid \forall i, \langle x_{F,i}, x_R \rangle \in \text{dp}\} \right),$$

the infimal resource satisfying all required functionalities. Lifting  $r_{\min}$  via  $\mathcal{P}$  gives a query  $\mathcal{P}(r_{\min}): \mathcal{P}(\text{DP}_A\{F, R\}) \rightarrow \mathcal{P}(R)$  from distributions on DPs to induced distributions on minimal resources. An inner confidence bound  $[\text{dp}_L, \text{dp}_U]$  yields  $[r_{\min}(\text{dp}_L), r_{\min}(\text{dp}_U)]$ , a bound on the minimal resource at the same probability. Currying  $\text{ev}$  over  $R^{\mathbb{N}}$  further lets  $r_{\min}$  take the required functionalities as extra inputs and stay quasi-measurable.

## V. NUMERICAL EXAMPLE: UNCERTAIN TASK-DRIVEN CO-DESIGN OF UNMANNED AERIAL VEHICLES

We instantiate the framework on a task-driven UAV delivery problem adapted from [9], [52]. The goal is to expose in one benchmark the gap between (i) deterministic co-design, (ii) interval uncertainty, and (iii) distributional uncertainty with adaptive decision-making. Section V-A gives the co-design model and decomposition; Section V-B solves the deterministic and interval baselines and clarifies how to read the optimistic/pessimistic envelope; Section V-C adds distributional uncertainty in the task profile and component specifications, computing distributions of optimal cost by Monte Carlo; Section V-D compares procedures differing in *which* choices may depend on *which* observations, in the language of Section IV-B.

### A. Task-driven unmanned aerial vehicle co-design

An UAV must complete delivery missions over its operational lifetime. A *task profile* comprises *number of delivery*

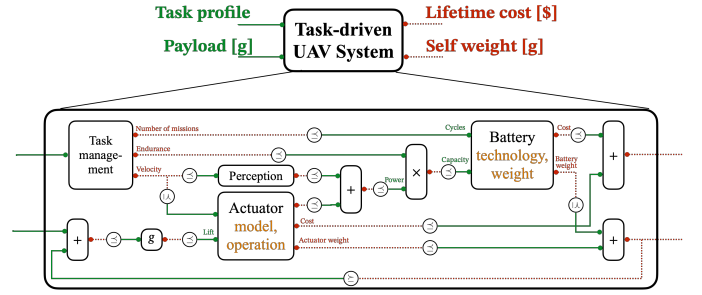


Fig. 4: Co-design diagram for a task-driven UAV, decomposed into functional components; designs carry **implementation** color. Red and green lines leaving the lower diagram are the four system-level **functionalities/resources** of the interface above.

*missions*, *distance coverage*, and *mission frequency*; given one, the objective is to maximize *payload* while minimizing *lifetime cost* and *self weight*. In co-design this is a system DP providing *task profile* and *payload* and requiring *lifetime cost* and *self weight* (Fig. 4, top), decomposed as in Fig. 4 (bottom) following [2]. For interpretability, explicit **implementations** appear only in *Actuation* (model, operating point) and *Battery* (technology, weight); co-design extends to richer architectures with minimal overhead [4], [6].

Recall that colors record the *role* a quantity plays at the interface where it appears, not an intrinsic property: *velocity* is a **functionality** the actuation component provides, hence green at the actuation interface, while it is a **resource** required by the task-management block, hence red there Fig. 4. This role reversal is performed by series composition.

*Task management*: For each *task profile*, this block derives intermediate requirements: *number of missions*, *endurance*, *velocity*.

*Perception*: The perception stack (sensor and software) is fixed, hence has no **implementation** choice; its *power* consumption increases with *velocity* [53].

*Actuation*: Three actuator **models** are considered (Table I), each with deterministic *weight*, *cost*, and *maximum velocity*. An **operation** decision allocates *power*  $P$  to *lift*  $L$  via  $P \geq p_0 + p_1 L^2$ , with  $p_0, p_1$  **specifications** of this DP. Here the *parameters* of design choices (some uncertain later) coincide with the DP **specifications**; in general they may differ, as long as each choice's parameters yield a component **specification**.

*Battery*: A battery design is a **technology** choice (Table I) plus a *battery weight* decision. Its DP is specified by *energy density*, *unit energy per cost*, and *cycle life* (also the technology parameters), fixing the trade-off between provided *capacity/cycles* and required *battery weight/total cost*, maintenance and replacement included.

### B. Deterministic and interval uncertainty models

We begin with deterministic parameters and free choice over all **implementations**. Fixing the *task profile*, we solve the **FixFunMinRes** query for the co-design model in Fig. 4 and project the resulting DP to *payload* versus *lifetime cost* for visualization (*payload* is *not* in the **functionality** requirement but part of the resulting trade-off, as noted in Remark 3). Fig. 5 reports the trade-off: larger payloads force higher lifetime cost, illustrating monotonicity of the feasible relation. For the interval baseline we perturb the nominal parameters of Table I

| Actuator | Mass [g] | Cost [\$] | Velocity [m/s] | $p_0$ [W] | $p_1$ [W/N <sup>2</sup> ] |
|----------|----------|-----------|----------------|-----------|---------------------------|
| $a_1$    | 50.0     | 50.0      | 3.0            | 1.0       | 2.0                       |
| $a_2$    | 100.0    | 100.0     | 3.0            | 2.0       | 1.5                       |
| $a_3$    | 150.0    | 150.0     | 3.0            | 3.0       | 1.5                       |

| Battery Technology | Energy density [Wh/kg] | Unit power per cost [Wh/\$] | Number of cycles |
|--------------------|------------------------|-----------------------------|------------------|
| NiMH               | 100.0                  | 3.41                        | 500              |
| NiH2               | 45.0                   | 10.50                       | 20,000           |
| LCO                | 195.0                  | 2.84                        | 750              |
| LMO                | 150.0                  | 2.84                        | 500              |
| NiCad              | 30.0                   | 7.50                        | 500              |
| SLA                | 30.0                   | 7.00                        | 500              |
| LiPo               | 150.0                  | 2.50                        | 600              |
| LFP                | 90.0                   | 1.50                        | 1,500            |

TABLE I: Deterministic parameters of component choices [9].

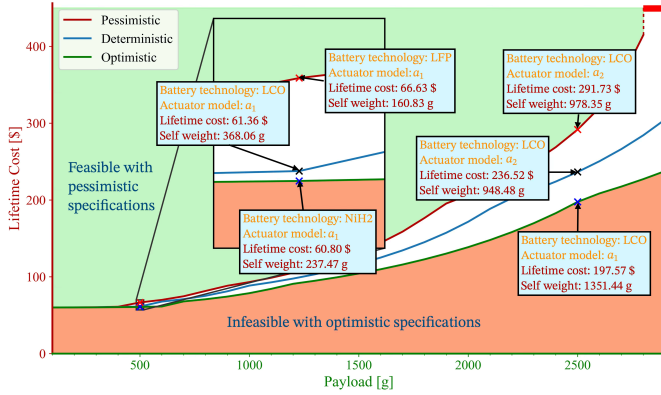


Fig. 5: Required *payload* versus minimal *lifetime cost* for a fixed *task profile*, deterministic and interval-uncertain. Pairs above the pessimistic curve (green) are robustly feasible, below the optimistic one (orange) infeasible even optimistically; between them, feasible for some realizations only. Labels give optimal *implementations* at some payloads; for large payloads the pessimistic model is infeasible, with minimal cost  $+\infty$ .

by  $\pm 5\%$  into optimistic and pessimistic variants and solve the same query for each. The envelope must be read with care: as the two problems are solved *independently*, it is a *wait-and-see* interpretation in which the *implementation* optimal for each realized instance may be chosen, so the optimal actuator/battery may differ between the curves. If the designer must instead commit *before* uncertainty is realized (here-and-now), the envelope does not characterize a fixed commitment: picking *actuator*  $a_1$  and *battery* LCO at payload 2500 g from the optimistic curve guarantees neither feasibility nor cost under the pessimistic realization, which may require *actuator*  $a_2$ . This motivates explicit distributions and staged policies.

### C. Distributional models and confidence bounds with all decisions after observing outcomes

We now add distributional uncertainty in (i) the task profile and (ii) the actuation and battery parameters. We first take a *maximally adaptive* benchmark in which all discrete *implementation* choices follow the observation of the realized parameters: in the language of Section IV, it is exactly the lifted union as *free choice after observation*. Section V-D then restricts it to realistic adaptivity levels.

a) *Uncertain task profile*: *Mission frequency* and *distance coverage* are deterministic; the *number of delivery missions* is random, modeled with an extra DP (Remark 6, Task in Fig. 8).

b) *Sample space and distributions*: Let  $\Omega$  be the product of the random parameters defining the task and all available choices:

$$\Omega = \Omega_T \times \Omega_B \times \Omega_A = (\Omega_N) \times \left( \prod_{b \in \mathcal{B}} \Omega_b \right) \times \left( \prod_{i \in \{1,2,3\}} \Omega_{a_i} \right),$$

with  $\Omega_N = \mathbb{N}$  the number of missions,  $\Omega_b$  the uncertain parameters of battery technology  $b \in \mathcal{B}$  (Table I), and  $\Omega_{a_i} = \mathbb{R}_{\geq 0}^2$ ,  $\omega_{a_i} = \langle p_{0,i}, p_{1,i} \rangle$ , those of actuator  $a_i$ .

Each uncertain scalar has an independent (truncated) Gaussian marginal, with mean the value in Table I and variance calibrated so its 90% confidence interval is  $\pm 5\%$ ;  $\mu$  is the product of all marginals.

c) *Random DPs for components*: To lighten notation, write  $\omega = \langle \omega_T, \omega_B, \omega_A \rangle$  and display only the coordinates relevant to each component. With  $\langle v, L \rangle$  the *velocity* and *lift* functionalities and  $\langle p, c, w \rangle$  the *power*, *cost*, and *weight* resources, the random performance of actuator model  $a_i$  is

$$\alpha_{a_i} : \Omega \rightarrow \text{DP}_A\{\mathbb{R}_{\geq 0}^2, \mathbb{R}_{\geq 0}^3\}, \omega \mapsto \left\{ \langle \langle v, L \rangle, \langle P, c, w \rangle \rangle \mid v \leq v_{a_i}, c_{a_i} \leq c, w_{a_i} \leq w, p_{0,i} + p_{1,i}L^2 \leq P \right\}.$$

Since  $\alpha_{a_i}$  depends on  $\omega$  only through  $\omega_{a_i} = \langle p_{0,i}, p_{1,i} \rangle$ , the sample-space decomposition required for constructive composition (Remark 8) applies.

As the benchmark lets the designer choose the actuator *after* observing  $\omega$ , actuation DP is the pointwise deterministic union

$$\alpha_A = \vee \circ \langle \alpha_{a_1}, \alpha_{a_2}, \alpha_{a_3} \rangle : \Omega \rightarrow \text{DP}_A\{\mathbb{R}_{\geq 0}^2, \mathbb{R}_{\geq 0}^3\}.$$

The battery is analogous, with  $\alpha_B$  the union over technologies after observing  $\omega_B$ . Following Remark 6, stochasticity in the *task profile* is modeled with a special DP (Task in Fig. 8) with one *resource* interface and trivial *functionality*: the singleton poset [3]. With  $d_r, \nu_r$  the required distance coverage and mission frequency and  $\omega_N \in \Omega_N = \mathbb{N}$  the random number of deliveries,

$$\alpha_T : \Omega \rightarrow \text{DP}_A\{\{\star\}, \mathbb{R}_{\geq 0}^2 \times \mathbb{N}\}, \omega \mapsto \{ \langle \star, \langle d, \nu, n \rangle \rangle \mid d \geq d_r, \nu \geq \nu_r, n \geq \omega_N \}.$$

d) *System-level random DP*: All remaining blocks of Fig. 4 are deterministic, embedded as constant random variables. As the decomposition of Remark 8 applies, the system random variable is explicit: with  $f$  composing the deterministic component DPs of Fig. 4 and the Task DP of Fig. 8,

$$\alpha_{\text{UAV}}^{\text{post}} = f \circ \langle \alpha_T, \alpha_B, \alpha_A, \dots \rangle. \quad (1)$$

Fixing the distribution  $\mu$  on  $\Omega$  yields the distributional uncertain design result  $\langle \mu, \alpha_{\text{UAV}}^{\text{post}} : \Omega \rightarrow \text{DP}_A\{\mathbb{R}_{\geq 0}^2, \mathbb{R}_{\geq 0}^3\} \rangle$ , with *payload* the functionality and *lifetime cost* the resource (projected for visualization).

e) *Inner confidence bounds*: Inner confidence bounds follow from monotonicity of the component models in their parameters. Endow  $\Omega$  with the product order: actuation is antitone in  $\langle p_{0,i}, p_{1,i} \rangle$  (larger values worsen feasibility) while key battery parameters (energy density, energy-per-cost) are monotone, so after taking opposites where needed the dependence is monotone and rectangular intervals  $[\omega_L, \omega_U] \subseteq \Omega$  induce DP intervals  $[\alpha(\omega_L), \alpha(\omega_U)]$  as inner bounds. By independence, each scalar interval at level  $\rho$  gives rectangle



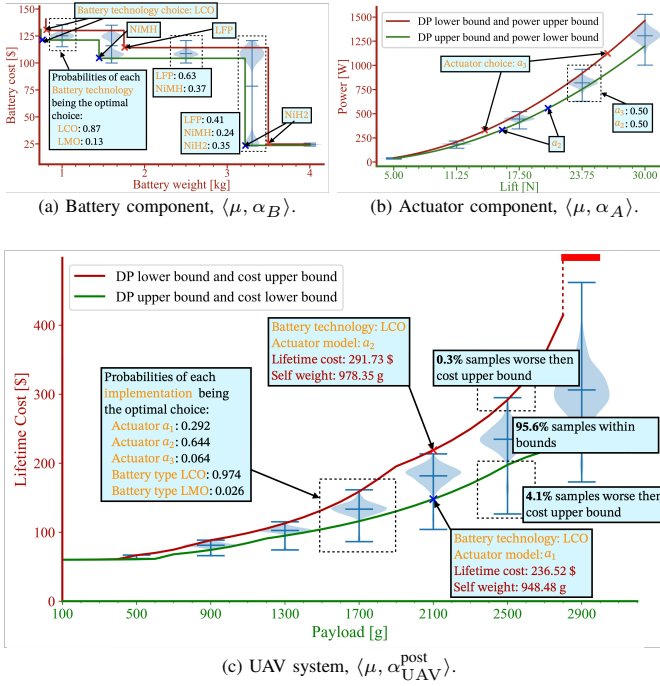


Fig. 6: Uncertain design results for the task-driven UAV and selected components, maximally adaptive benchmark. Upper and lower curves are inner confidence bounds; violin plots are distributions of minimal **resource** via Example 4. Some distributions are annotated with the probability of each **implementation** being optimal and share of samples inside/outside the bound.

probability  $\rho^K$  for  $K$  uncertain scalars, so an overall level  $p$  needs  $\rho = p^{1/K}$ .

f) *Numerical design results:* To visualize results we apply the observation of Example 4, giving distributions on one **resource** with all other **functionalities** and **resources** fixed, estimated by Monte Carlo sampling of  $\omega$  from  $\mu$ ; bounds are curves.

Fig. 6a shows the **battery weight/battery cost** trade-off when requesting **150 Wh** and **1000 cycles**. The two bounds come from discrete choices, whereas the distributional result also captures (a) rare cases where an **implementation** off the bounds is optimal (**LMO**, first annotated distribution); (b) mixtures of the bounding **implementations** when these differ (**LFP** and **NiMH**, second); and (c) the gradual take-over of a new **implementation** as **weight** increases (third), in contrast to the abrupt jumps and large gaps of the bounds. Fig. 6b shows the actuator trade-off projected onto **lift** and **power**, Fig. 6c the composed UAV system: components and system share *the same language* and visualization (framework's modularity). The  $\pm 5\%$  per-parameter bounds give  $\rho = 0.9$ , hence for  $K = 30$  scalars a composed level of only  $p = \rho^{30} \approx 0.042$ . The product construction is thus conservative while the distributional composition is *exact*: empirically over 90 % of realized minimal **lifetime costs** fall inside the composed bound, above the 4.2 % guarantee (Fig. 6c).

#### D. Adaptive decision processes and design results

The maximally adaptive benchmark postpones every discrete **implementation** choice until the parameters are observed; practical processes restrict what may be postponed and what must be made first. We compare three adaptivity levels, all

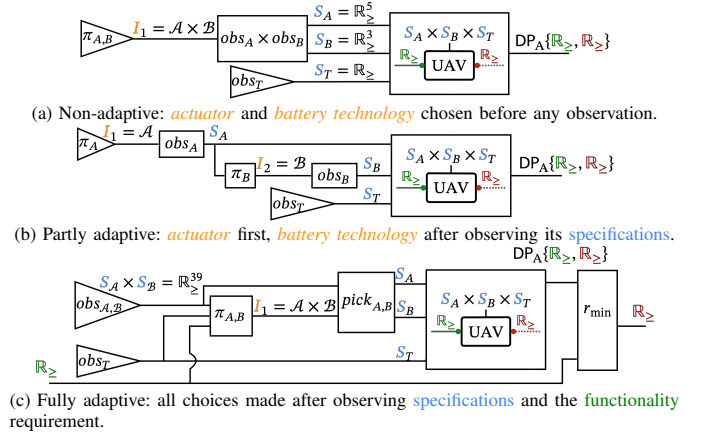


Fig. 7: Design processes for the UAV with different adaptivity levels.

allowing fully adaptive *recourse* allocations (actuator operating point, battery weight) within components, differing only in how the discrete actuator and battery choices may depend on observations.

a) *Non-adaptive:* **Actuator** and **battery technology** are chosen before any observation, while actuator **operation** and battery **weight** are optimized after uncertainty is realized. In Fig. 7a the policy  $\pi_{A,B}$  is a kernel with no input returning a distribution over  $\mathcal{A} \times \mathcal{B}$ : a possibly randomized commitment to an actuator–battery pair. Observation kernels then give distributions over the realized specifications of the chosen components, e.g.  $\text{obs}_A: \mathcal{A} \rightarrow \mathcal{P}(\mathbb{R}_{\geq}^5)$ ,  $a_i \mapsto \langle \mu, \alpha_{a_i}: \omega \mapsto \langle m_{a_i}, c_{a_i}, v_{a_i}, p_{0,i}, p_{1,i} \rangle \rangle$ , with  $p_{0,i}, p_{1,i}$  coordinates of  $\omega$ ;  $\text{obs}_B, \text{obs}_T$  are analogous. The system model  $a_{UAV}: \mathcal{S}_A \times \mathcal{S}_B \times \mathcal{S}_T \rightarrow \mathcal{P}(\text{DP}_A\{F, R\})$  then returns a distribution over system-level DPs. Here  $a_{UAV}$  returns a delta distribution (a constant random variable) over  $\text{DP}_A\{F, R\}$  per **specification**: with  $f$  as in Eq. (1) and  $g_T, g_B, g_A$  returning deterministic component DPs for the corresponding **specifications** (notation-wise omitting other deterministic components),  $a_{UAV}(s_A, s_B, s_T) = \langle \mu, \omega \mapsto f(g_T(s_T), g_B(s_B), g_A(s_A), \dots) \rangle$ . Decisions inside the DP (intermediate **resource/functionality** pairs, actuator operation, battery weight) are made with all outcomes observed, and we read off the random minimal **lifetime cost** per **payload** by post-composing  $r_{\min}$  as a deterministic kernel.

b) *Partly adaptive:* The **actuator** is chosen before observing its parameters, the **battery technology** after observing the realized actuator **specifications**. In Fig. 7b,  $\pi_A$  commits a (randomized) choice with no observation while  $\pi_B$  maps observed actuator specifications to a distribution over technologies. This is a three-stage re-parameterization: the third-stage choice depends on a second-stage observation, which in turn depends on the first-stage choice (Fig. 8).

c) *Fully adaptive:* The parameters of *all* actuator models and battery technologies are observed before selecting. In Fig. 7c,  $\text{obs}_{A,B}$  outputs a distribution over all parameters of Table I, from which  $\pi_{A,B}$  picks a pair given the observed task number and **payload**. This is the policy-based counterpart of the lifted-union benchmark: always selecting the optimal pair per realized parameter and **payload** reproduces the post-observation semantics.

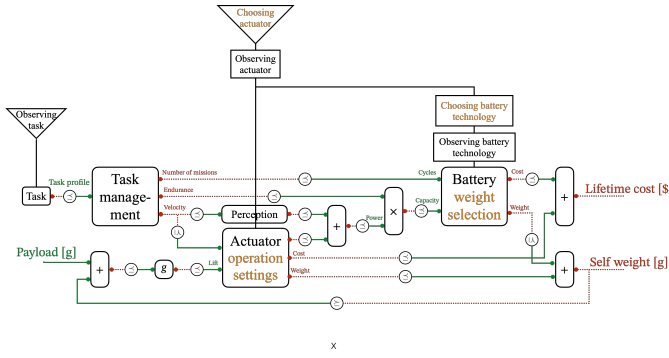


Fig. 8: Partly adaptive decision process combined with the co-design decomposition of the task-driven UAV.

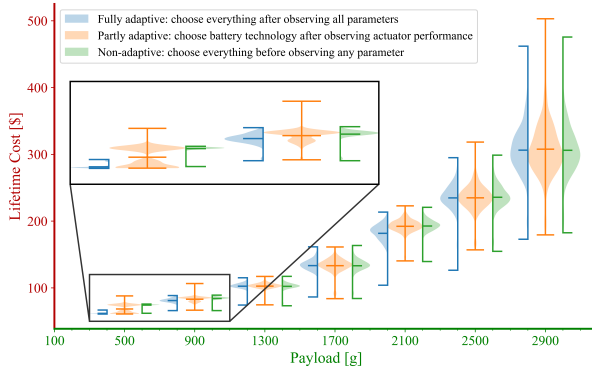


Fig. 9: Design results for three adaptivity levels: distributions of minimal lifetime cost induced by each design process at the given payloads.

d) *Policies used for comparison:* The framework allows general, including randomized, policies, which matter under chance constraints or explicit risk measures [33]. We let the policies in Figures 7a and 7b depend on *payload* and use a simple deterministic template: *given the observations, choose the implementation most likely to be optimal for minimal lifetime cost*, estimating from Monte Carlo samples the probability that each implementation attains the minimum and taking the maximum a posteriori option. This approximately optimizes expected minimal cost under each information structure.

e) *Results and interpretation:* Fig. 9 reports the distributions of minimal lifetime cost for several payloads. Improved adaptivity substantially reduces expected lifetime cost in some payload regimes (near feasibility boundaries, where technology switching matters most), while the upper tails may not improve and can even worsen. This is the distinction between “average” performance and low-probability, high-impact outcomes: exactly the trade-off interval bounds cannot quantify and distributional co-design makes explicit.

## VI. CONCLUSIONS AND FUTURE WORK

We developed a compositional framework for co-design under *distributional* uncertainty with explicit *adaptive* multi-stage decision-making. Building on monotone co-design, we used Quasi Measurable Space/Quasi Universal Space to define distributions over design problems that avoid measurability pathologies: DP compositions (series, parallel, feedback, union, intersection) remain measurable and lift to those distributions, feasibility probabilities are computable, stochasticity in *functionality/resources* are representable, and deterministic

theory becomes a special case. We introduced adaptive design processes via Markov-kernel re-parameterizations, supporting uncountable specification spaces, dependence between specifications and choices, and risk-aware objectives; and formalized *queries* and *observations* as the interface between model and engineering questions, including probability-of-feasibility, confidence bounds, and minimal-resource observations, all illustrated on an uncertain task-driven UAV study.

Several directions remain. Algorithmically, we rely on Monte Carlo estimation; a natural next step is *compositional* estimators and probability bounds exploiting the co-design graph, including variance-reduced sampling, concentration certificates, and deterministic bounds under monotonicity. On the modeling side, risk measures and constraints (chance constraints, CVaR-type objectives, distributionally robust variants) can enter the query layer directly, alongside policy synthesis under partial observations, and the framework suggests an end-to-end learning loop: infer and update specification distributions from data, propagate uncertainty compositionally, and optimize adaptive policies at the system level. Finally, given [46], an end-to-end formalization of further uncertainty layers (e.g. ambiguity sets [10]) is of great interest.

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Award of Excellence as a Teaching Assistant at MIT.



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## VII. APPENDIX

### A. Reading guide: what the categorical language buys

This appendix is a reference. It is more useful read as an answer to a single question: *why does a paper about design trade-offs end up talking about categories at all?* This subsection gives that answer by walking through the paper's development and naming, at each step, the structure that is doing the work in the background, together with the definition below that makes it precise. None of it is needed to follow the paper; it is offered because the recurring pattern is, we think, the most portable part of the contribution.

The short version is this. The paper's objects of study are not numbers or signals but *relations between posets*, and later *distributions over such relations*. These are higher-order objects, and the two questions one always asks of them, i.e., *can I interconnect them?* and *can I put a probability measure on them?*, have standard answers only for much simpler objects. Category theory is where the vocabulary for "things that compose in series and in parallel, that admit feedback, and that carry extra structure" already exists, and, as Counterexample 1 shows, it is also where the obstruction to the naive approach is visible.

1) *Deterministic co-design is a traced symmetric monoidal category*: Take posets as objects (the interfaces at which components meet) and DPs as morphisms (the processes connecting them). Then the series composition of Definition 9 is exactly the composition of a *category* (Definition 29), the parallel composition is a *monoidal product* (Definition 34), the fact that the order of parallel components is immaterial is *symmetry* (Definition 35), and feedback is a *trace* (Definition 37). This is the content of Remark 1. The payoff is concrete: a co-design diagram such as Fig. 4 denotes one well-defined DP, independently of the order in which its boxes are wired together, and this is a theorem about traced symmetric monoidal categories (*coherence*), not an assumption we make about diagrams. Because  $\text{DP}\{F, R\}$  additionally carries an order, the structure is *enriched* in posets (Definition 43): hom-sets are not bare sets but ordered ones, which is what makes "a better DP" meaningful and monotonicity composable.

2) *Design choices break strictness, and that is informative*: Lifting the operations to MDPIs (Definition 11) almost repeats the story, but not quite: products of implementation sets are associative only up to a canonical bijection. The honest statement is therefore bicategorical, with those bijections recorded as 2-cells (Definition 42), which is Remark 2. This is worth noticing rather than hiding, because it is the first appearance of a distinction the paper later depends on: between a design result and a *re-parameterization* of the process that produced it.

3) *Function spaces are the whole difficulty*: Counterexample 1 says no sigma algebra on  $\text{DP}\{F, R\}$  supports the operations we need. Stated categorically, the category of measurable spaces lacks the *exponential objects* (spaces of maps) that the construction requires; this is the same defect that blocks higher-order probability. QMSs are engineered to repair exactly this: together with products and a terminal object they form a *Cartesian* closed category (Definition 36), which is

Remark 5. Closedness is not decoration here. It is what makes *currying* available (Lemma 1), and currying is what turns the evaluation map of Theorem 4 into the query  $\text{pr}$  and the observation  $r_{\min}$  of Example 4, i.e., into observations that return *maps and sets* rather than only numbers. Of everything in this appendix, this is the one structural fact the paper could not do without.

4) *Distributions form a monad, and adaptivity is its Kleisli category*:  $\mathcal{P}$  of Definition 19 acts on spaces and on maps, and Lemma 2, which lifts deterministic maps to distributions, is precisely the statement that it is a *functor* (Definition 32); its unit (a point becomes a Dirac distribution) and multiplication (a distribution over distributions averages out) are *natural transformations* (Definition 33) satisfying the *monad laws* (Definition 38). Markov kernels in the sense of Definition 20 are then morphisms of the associated *Kleisli category* (Definition 41), and their composition is Kleisli composition. Adding the structural maps that copy and discard makes the monad *strong* and, on QUSs, *commutative* (Definitions 39 and 40), so the Kleisli category is symmetric monoidal. That is Remark 12, and it is what licenses Fig. 7: an arbitrary wiring of policies and observations denotes a single well-defined kernel, so the reader may draw the information structure of a design process and trust that it means something. Commutativity is the technical reason independent randomness may be sampled in either order, which is what Remark 8 exploits when it decomposes the sample space.

5) *The uncertain theory is the deterministic theory, enriched*: Theorem 3 says the co-design operations survive the passage to distributions. Categorically, the objects are unchanged and only the hom-objects have grown: uncertain co-design is a symmetric monoidal category *enriched* in QUS (Definition 44), which is Remark 7. This is the sense in which the paper claims to extend rather than replace monotone co-design: the deterministic theory sits inside the uncertain one as the sub-theory of point-mass distributions, via the constant random variables of Definition 21. Layering the re-parameterizations of Definition 25 on top gives the 2-categorical picture of Remark 11, with uncertain co-design problems as 1-cells and design processes as 2-cells, and it is this two-level view that [46] generalizes to other kinds of uncertainty.

*Where to look*. Definitions 29 and 31 to 33 are the basic vocabulary and suffice for Remark 1. Definitions 34, 35 and 37 formalize parallel composition and feedback. Definition 36 underlies QMSs and every query and observation. Definitions 38 to 41 formalize distributions and adaptivity. Definitions 42 to 44 are needed only for the two remarks that describe the overall architecture.

Finally, a caveat that we mean literally. A reader who skips this appendix entirely, and every remark marked (CT), loses no definition, no theorem, and no step of any proof: the main text is written in ordinary set- and measure-theoretic language throughout. What is lost is only the explanation of *why* the constructions take the shape they do, and why we expect the same shape to recur when other kinds of uncertainty are added to other compositional design formalisms.

## B. Review of category theory

We collect here the category-theoretic notions used throughout the paper. The language is adapted for readers without a heavy category theory background: whenever possible, we relate each concept to familiar objects such as sets, maps, state spaces, and the interconnection of subsystems.

1) *Categories, functors, and natural transformations*: A category is the *minimal* algebraic structure needed to talk about *composing systems with specific interfaces (between specific spaces)* (e.g., composing dynamic systems with input/outputs). The only two things a category insists on are that maps compose *associatively* (chaining subsystems does not depend on how we bracket the chain) and that every interface carries a *do-nothing* identity process.

**Definition 29** (Category). A *category*  $\mathcal{C}$  consists of the following constituents.

- 1) A set  $\text{Ob}(\mathcal{C})$  of *objects*  $A, B, C, \dots$
- 2) For every ordered pair of objects  $A, B$ , a set  $\mathcal{C}(A, B)$  of *morphisms* (or *arrows*) from  $A$  to  $B$ ; we write  $f: A \rightarrow B$  for  $f \in \mathcal{C}(A, B)$ , and call  $A$  its *domain* and  $B$  its *codomain*.
- 3) For every triple of objects  $A, B, C$ , a *composition* map  $\circ: \mathcal{C}(A, B) \times \mathcal{C}(B, C) \rightarrow \mathcal{C}(A, C)$ , written  $\langle f, g \rangle \mapsto f \circ g$ .
- 4) For every object  $A$ , an *identity* morphism  $\text{id}_A \in \mathcal{C}(A, A)$ .

These are subject to the following conditions, for all composable  $f: A \rightarrow B, g: B \rightarrow C, h: C \rightarrow D$ .

- 1) *Associativity*:  $(f \circ g) \circ h = f \circ (g \circ h)$ .
- 2) *Left and right unitality*:  $\text{id}_A \circ f = f = f \circ \text{id}_B$ .

**Definition 30** (Product category). Given categories  $\mathcal{C}$  and  $\mathcal{D}$ , their product  $\mathcal{C} \times \mathcal{D}$  is defined by:

- 1)  $\text{Ob}(\mathcal{C} \times \mathcal{D}) = \text{Ob}(\mathcal{C}) \times \text{Ob}(\mathcal{D})$  (the Cartesian product of sets).
- 2)  $\mathcal{C} \times \mathcal{D}(\langle A, B \rangle, \langle C, D \rangle) = \mathcal{C}(A, C) \times \mathcal{D}(B, D)$  (also the Cartesian product).
- 3) Composition maps as Cartesian products of composition maps of  $\mathcal{C}$  and  $\mathcal{D}$ .
- 4) For every  $\langle A, B \rangle \in \text{Ob}(\mathcal{C} \times \mathcal{D})$ ,  $\text{id}_{\langle A, B \rangle} = \langle \text{id}_A, \text{id}_B \rangle$ .

**Remark 15.** Category theory can serve as the basic construction of mathematics, just as set theory commonly does. In those situations, one would not mention “set” in the definition of categories, but a more general “collections” of objects or call objects “types”, and categories whose objects form a set are called “small categories”. In this introductory section, we stick to the language of set theory.

It is common to use *commutative diagrams* rather than equations to express conditions graphically in category theory. In those diagrams, morphisms are drawn as arrows between objects, and directed paths with the same origin and end objects are constrained to be equal (the reason for calling them “commutative”). The associativity condition in Definition 29 makes those paths well defined without ambiguity. For instance, the path from  $A$  to  $D$  in the following simple diagram:

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

can be interpreted as  $(f \circ g) \circ h$  or  $f \circ (g \circ h)$ , with associativity confirming that they are the same thing. As an example of a commutative diagram, the identity conditions in Definition 29 can be equivalently visualized by:

$$\begin{array}{ccccc} A & \xrightarrow{\text{id}_A} & A & \xleftarrow{\text{id}_B} & B \\ & \searrow f & \downarrow f & \swarrow f & \\ & & B & & \end{array}$$

or

$$\begin{array}{ccc} \text{id}_A & & \text{id}_B \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \end{array}$$

Pay attention to the different notation of diagrams in this paper, such as Fig. 1, where boxes are morphisms and wires are objects. Wires being morphisms (commutative diagrams) is convenient for conditions and research about category theory itself, while boxes being morphisms is often convenient in applying category into other domains (e.g., compositional probability theory and co-design).

The prototypical example is *Set*, whose objects are sets and whose morphisms are functions. Another example is the category of linear dynamic systems without constraints, with natural numbers (or Euclidean spaces of the corresponding dimension) as objects and linear dynamic systems with corresponding input/output dimensions as morphisms. Composition is then the normal system composition by feeding the output of one system to the input of another.

Two objects are “the same up to relabeling” when there is a morphism between them that can be undone, exactly as an invertible change of coordinates identifies two state spaces.

**Definition 31** (Inverses and isomorphisms). Let  $\mathcal{C}$  be a category and  $f: A \rightarrow B$  a morphism.

- 1) A morphism  $g: B \rightarrow A$  is an *inverse* of  $f$  if  $f \circ g = \text{id}_A$  and  $g \circ f = \text{id}_B$ .
- 2)  $f$  is an *isomorphism* if it admits an inverse; we then write  $f^{-1}$  for its (necessarily unique) inverse and say  $A$  and  $B$  are *isomorphic*, denoted  $A \cong B$  or  $f: A \xrightarrow{\cong} B$ .

A functor is a *structure-preserving map between categories*: it relabels objects and morphisms while respecting how they compose.

**Definition 32** (Functor). Let  $\mathcal{C}$  and  $\mathcal{D}$  be categories. A *functor*  $F: \mathcal{C} \rightarrow \mathcal{D}$  consists of the following constituents.

- 1) An *object map* sending each  $A \in \text{Ob}(\mathcal{C})$  to  $FA \in \text{Ob}(\mathcal{D})$ .
- 2) A *morphism map* sending each  $f: A \rightarrow B$  in  $\mathcal{C}$  to  $Ff: FA \rightarrow FB$  in  $\mathcal{D}$ .

These are subject to the following conditions.

- 1) *Preservation of composition*:  $F(f \circ g) = Ff \circ Fg$  for all composable  $f, g$ .
- 2) *Preservation of identities*:  $F(\text{id}_A) = \text{id}_{FA}$  for all objects  $A$ .

The identity functor for  $\mathcal{C}$ , denoted as  $\text{id}_{\mathcal{C}}$ , has all the constituents as identity maps between the corresponding sets. If we regard functors themselves as maps, a natural transformation is a *map between two functors*: a coherent, object-

indexed family of morphisms that transforms the output of one functor into that of another.

**Definition 33** (Natural transformation). Let  $F, G: \mathcal{C} \rightarrow \mathcal{D}$  be functors. A *natural transformation*  $\eta: F \Rightarrow G$  consists of the following constituent.

- 1) A family of morphisms  $\{\eta_A: FA \rightarrow GA\}_{A \in \text{Ob}(\mathcal{C})}$  in  $\mathcal{D}$ , called its *components*.

It is subject to the following condition.

- 1) *Naturality*: for every  $f: A \rightarrow B$  in  $\mathcal{C}$ ,  $Ff \circ \eta_B = \eta_A \circ Gf$ .

A natural transformation whose components are all isomorphisms is a *natural isomorphism*. When convenient we use the morphism notion to talk about natural transformations, for instance “the natural transformation  $\eta_A: FA \rightarrow GA$ ”.

The naturality condition graphically is the commutativity of the *naturality square*:

$$\begin{array}{ccc} FA & \xrightarrow{Ff} & FB \\ \eta_A \downarrow & & \downarrow \eta_B \\ GA & \xrightarrow{Gf} & GB \end{array}$$

**2) Monoidal and symmetric monoidal categories:** So far, a category only lets us *chain* systems (series composition). A monoidal structure adds the ability to place systems *side by side* (parallel composition), together with a trivial “empty” system that acts as a unit. This is precisely the parallel composition of DPs in Definition 9. The subtlety is that regrouping  $(A \otimes B) \otimes C$  into  $A \otimes (B \otimes C)$ , or padding with the unit, need not be equal *on the nose*; instead, they are related by fixed isomorphisms that must fit together coherently.

**Definition 34** (Monoidal category). A *monoidal category* extends a category  $\mathcal{C}$  (Definition 29) with the following constituents.

- 1) A functor  $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ , the *tensor product*.
- 2) A distinguished object  $I$ , the *unit*.
- 3) A natural isomorphism  $\alpha_{A,B,C}: (A \otimes B) \otimes C \xrightarrow{\cong} A \otimes (B \otimes C)$ , the *associator*.
- 4) Natural isomorphisms  $\lambda_A: I \otimes A \xrightarrow{\cong} A$  and  $\rho_A: A \otimes I \xrightarrow{\cong} A$ , the *left and right unitors*.

These are subject to the following coherence conditions.

- 1) *Pentagon*: the two ways of reassociating  $((A \otimes B) \otimes C) \otimes D$  into  $A \otimes (B \otimes (C \otimes D))$  agree.

$$\begin{array}{ccc} & (A \otimes B) \otimes (C \otimes D) & \\ \alpha \swarrow & & \searrow \alpha \\ ((A \otimes B) \otimes C) \otimes D & & A \otimes (B \otimes (C \otimes D)) \\ \alpha \otimes \text{id} \downarrow & & \uparrow \text{id} \otimes \alpha \\ (A \otimes (B \otimes C)) \otimes D & \xrightarrow{\alpha} & A \otimes ((B \otimes C) \otimes D) \end{array}$$

- 2) *Triangle*: reassociating  $(A \otimes I) \otimes B$  and then applying the right unitor equals applying the left unitor after reassociation.

$$\begin{array}{ccc} (A \otimes I) \otimes B & \xrightarrow{\alpha} & A \otimes (I \otimes B) \\ \rho \otimes \text{id} \searrow & & \swarrow \text{id} \otimes \lambda \\ & A \otimes B & \end{array}$$

Parallel composition of physical systems often does not care about left-to-right order: running  $A$  alongside  $B$  is the same as running  $B$  alongside  $A$ . A symmetry isomorphism makes this precise while remaining coherent with the associator and unitors of Definition 34.

**Definition 35** (Symmetric monoidal category). A *symmetric monoidal category* extends a monoidal category with the following constituent.

- 1) A natural isomorphism  $\sigma_{A,B}: A \otimes B \xrightarrow{\cong} B \otimes A$ , called the *symmetry* (or *braiding*).

It is subject to the following conditions.

- 1) *Involution*:  $\sigma_{A,B} \circ \sigma_{B,A} = \text{id}_{A \otimes B}$ .
- 2) *Hexagon*:  $\sigma$  is compatible with the associator  $\alpha$ , i.e., the hexagon diagram below commutes.

$$\begin{array}{ccccc} & & A \otimes (B \otimes C) & \xrightarrow{\sigma} & (B \otimes C) \otimes A \\ & \alpha \swarrow & & & \searrow \alpha \\ (A \otimes B) \otimes C & & & & B \otimes (C \otimes A) \\ \sigma \otimes \text{id} \searrow & & & & \swarrow \text{id} \otimes \sigma \\ & (B \otimes A) \otimes C & \xrightarrow{\alpha} & B \otimes (A \otimes C) & \end{array}$$

**3) Closed and traced structure:** Two further structures capture familiar operations. A Cartesian closed category has the monoidal product as the genuine *categorical product* (it uniquely bundles interfaces and systems), a *terminal object* (a “one point” interface into which there’s a unique special system from each other interface), and *internal homs*  $[A, B]$  that turn “spaces of systems  $f: A \rightarrow B$ ” into objects of the category itself, exactly as Set turns function spaces  $B^A$  into sets. This is the categorical home of currying.

**Definition 36** (Cartesian closed category). A *Cartesian closed category* is a symmetric monoidal category  $\mathcal{C}$  with the following extra constituents.

- 1) For each pair of objects  $A$  and  $B$ , projections  $\pi_1: A \otimes B \rightarrow A$  and  $\pi_2: A \otimes B \rightarrow B$ .
- 2) For every pair  $A, B$ , an *internal hom* (or *exponential*) object  $[A, B]$  (or  $B^A$ ) with an evaluation morphism  $\text{ev}_{A,B}: [A, B] \otimes A \rightarrow B$ .

These are subject to the following universal properties.

- 1) *Monoidal product is the categorical product*: for all  $f: C \rightarrow A$  and  $g: C \rightarrow B$  there is a unique  $\langle f, g \rangle: C \rightarrow A \otimes B$  with  $\langle f, g \rangle \circ \pi_1 = f$  and  $\langle f, g \rangle \circ \pi_2 = g$ .

$$\begin{array}{ccccc} & & C & & \\ f \swarrow & & \langle f, g \rangle & & \searrow g \\ A & \xleftarrow{\pi_1} & A \otimes B & \xrightarrow{\pi_2} & B \end{array}$$

- 2) *Monoidal unit as the terminal object*: for every object  $C$  there is a unique morphism  $!_C: C \rightarrow I$ .
- 3) *Exponentials (currying)*: for every  $h: C \otimes A \rightarrow B$  there is a unique  $\Lambda h: C \rightarrow [A, B]$  with  $(\Lambda h \otimes \text{id}_A) \circ \text{ev}_{A,B} = h$ . Intuitively, evaluating the first and the second coordinates gives the same result as evaluating both at the same time.

$$\begin{array}{ccc} [A, B] \otimes A & \xrightarrow{\text{ev}_{A,B}} & B \\ \Lambda h \otimes \text{id}_A \uparrow & & \uparrow h \\ C \otimes A & & \end{array}$$



A trace models *feedback*: it takes a system with an input/output pair of the same type and closes that channel into a loop. This is exactly the feedback operation on DPs in Definition 9, and it is why the co-design category is *traced*.

**Definition 37** (Traced symmetric monoidal category). A *traced symmetric monoidal category* extends a symmetric monoidal category with the following constituent.

- 1) A family of *trace operators*  $\text{Tr}_{A,B}^U: \mathcal{C}(A \otimes U, B \otimes U) \rightarrow \mathcal{C}(A, B)$ , indexed by objects  $A, B$  and the *feedback object*  $U$ , closing the loop on the  $U$ -component.

It is subject to the following axioms.

- 1) *Naturality in A*: for every  $f: A \otimes U \rightarrow B \otimes U$  and  $g: A' \rightarrow A$ ,

$$\text{Tr}_{A',B}^U((g \otimes \text{id}_U) \circ f) = g \circ \text{Tr}_{A,B}^U(f).$$

- 2) *Naturality in B*: for every  $f: A \otimes U \rightarrow B \otimes U$  and  $h: B \rightarrow B'$ ,

$$\text{Tr}_{A,B'}^U(f \circ (h \otimes \text{id}_U)) = \text{Tr}_{A,B}^U(f) \circ h.$$

- 3) *Dinaturality in U*: for every  $f: A \otimes U' \rightarrow B \otimes U$  and  $v: U \rightarrow U'$ ,

$$\text{Tr}_{A,B}^U((\text{id}_A \otimes v) \circ f) = \text{Tr}_{A,B}^{U'}(f \circ (\text{id}_B \otimes v)).$$

- 4) *Vanishing I*:  $\text{Tr}_{A,B}^U(f) = \rho_A^{-1} \circ f \circ \rho_B$  (tracing the unit does nothing up to unitors).

- 5) *Vanishing II*:

$$\text{Tr}_{A,B}^{U \otimes V}(\alpha_{A,U,V}^{-1} \circ f \circ \alpha_{B,U,V}) = \text{Tr}_{A,B}^U(\text{Tr}_{A \otimes U, B \otimes U}^V(f))$$

(a compound loop equals nested loops).

- 6) *Superposing*: for every  $f: A \otimes U \rightarrow B \otimes U$ ,

$$\text{Tr}_{C \otimes A, C \otimes B}^U(\alpha_{C,A,U} \circ (\text{id}_C \otimes f) \circ \alpha_{C,B,U}^{-1}) = \text{id}_C \otimes \text{Tr}_{A,B}^U(f)$$

(tracing with one more identity morphism does nothing).

- 7) *Yanking*:  $\text{Tr}_{U,U}^U(\sigma_{U,U}) = \text{id}_U$ .

Conditions in Definition 37 are hard to visualize with commutative diagrams. Interested readers are referred to [3] for visualizations with the morphisms-as-boxes language.

**4) Monads and their Kleisli categories:** Monads are the categorical tool for modeling extra *structures* on spaces and the corresponding *effects* layered on top of plain systems, most importantly the passage from deterministic maps to *probabilistic* ones. The functor  $T$  assigns to each space  $A$  a space  $TA$  of “ $T$ -decorated” values (e.g., distributions, robust sets, intervals, etc., over  $A$ ); the unit  $\eta$  embeds a plain value as a trivial (delta distribution, singleton set, singleton interval) effect; and the multiplication  $\mu$  flattens a doubly decorated value (a distribution over distributions) into a single one. The distributions-over-DPs construction in this paper is exactly by a monad on the category QUS.

**Definition 38** (Monad). A *monad* on a category  $\mathcal{C}$  consists of the following constituents.

- 1) An functor  $T: \mathcal{C} \rightarrow \mathcal{C}$  (Definition 32).
- 2) A natural transformation  $\eta: \text{id}_{\mathcal{C}} \Rightarrow T$ , the *unit*.
- 3) A natural transformation  $\mu: T^2 \Rightarrow T$ , the *multiplication*.

These are subject to the following conditions.

- 1) *Associativity*:  $T\mu_A \circ \mu_A = \mu_{TA} \circ \mu_A$ .

$$\begin{array}{ccc} T^3 A & \xrightarrow{T\mu_A} & T^2 A \\ \mu_{TA} \downarrow & & \downarrow \mu_A \\ T^2 A & \xrightarrow{\mu_A} & TA \end{array}$$

- 2) *Left and right unit laws*:  $\eta_{TA} \circ \mu_A = \text{id}_{TA} = T\eta_A \circ \mu_A$ .

$$\begin{array}{ccccc} TA & \xrightarrow{\eta_{TA}} & T^2 A & \xleftarrow{T\eta_A} & TA \\ \text{id}_{TA} \searrow & & \downarrow \mu_A & & \swarrow \text{id}_{TA} \\ & & TA & & \end{array}$$

To combine effects with parallel composition we need the monad to interact with the tensor product of Definition 34. A strength lets us pull a “context”  $A$  into a  $T$ -decorated value on  $B$ , i.e., broadcast side information into the effect, or take the product distribution with a delta distribution.

**Definition 39** (Strong monad). A *strong monad* on a monoidal category (Definition 34) is a monad  $(T, \eta, \mu)$  (Definition 38) with the following additional constituent.

- 1) A natural transformation  $t_{A,B}: A \otimes TB \rightarrow T(A \otimes B)$ , the (left) *tensorial strength*.

It is subject to the following coherence conditions.

- 1) *Unitality*:  $t_{I,B} \circ T\lambda_B = \lambda_{TB}$ .

- 2) *Associativity*:  $t_{A \otimes B, C}$  agrees with  $t_{A, B \otimes C}$  up to the associator  $\alpha$ .

$$\begin{array}{ccc} (A \otimes B) \otimes TC & \xrightarrow{t_{A \otimes B, C}} & T((A \otimes B) \otimes C) \\ \alpha_{A,B,TC} \downarrow & & \downarrow T\alpha_{A,B,C}^{-1} \\ A \otimes (B \otimes TC) & & T(A \otimes (B \otimes C)) \\ \text{id} \otimes t_{B,C} \downarrow & \nearrow t_{A,B \otimes C} & \\ A \otimes T(B \otimes C) & & \end{array}$$

- 3) *Compatibility with  $\eta$* :  $(\text{id}_A \otimes \eta_B) \circ t_{A,B} = \eta_{A \otimes B}$ .

$$\begin{array}{ccc} A \otimes B & \xrightarrow{\text{id} \otimes \eta_B} & A \otimes TB \\ \eta_{A \otimes B} \searrow & & \downarrow t_{A,B} \\ & & T(A \otimes B) \end{array}$$

- 4) *Compatibility with  $\mu$* :  $(\text{id}_A \otimes \mu_B) \circ t_{A,B} = t_{A, TB} \circ T t_{A,B} \circ \mu_{A \otimes B}$ .

$$\begin{array}{ccc} A \otimes T^2 B & \xrightarrow{t_{A, TB}} & T(A \otimes TB) \xrightarrow{T t_{A,B}} T^2(A \otimes B) \\ \text{id} \otimes \mu_B \downarrow & & \downarrow \mu_{A \otimes B} \\ A \otimes TB & \xrightarrow{t_{A,B}} & T(A \otimes B) \end{array}$$

For symmetric monoidal categories, a strength on one side induces one on the other; a monad is commutative when the two agree, so that broadcasting context from the left and from the right give the same result. Probabilistically, this says that independent effects can be combined in either order, i.e., a Fubini-type property for the product of distributions.

**Definition 40** (Commutative monad). Let  $(T, \eta, \mu, t)$  be a strong monad (Definition 39) on a symmetric monoidal category (Definition 35), with (left) strength  $t_{A,B}: A \otimes TB \rightarrow T(A \otimes B)$ . Define the *right strength*  $t'_{A,B}: TA \otimes B \rightarrow$

$T(A \otimes B)$  by  $t'_{A,B} = \sigma_{TA,B} \circ t_{B,A} \circ T\sigma_{B,A}$ , and the two induced maps  $TA \otimes TB \rightarrow T(A \otimes B)$ ,

$$\begin{aligned}\ell_{A,B} &= t_{TA,B} \circ Tt'_{A,B} \circ \mu_{A \otimes B}, \\ r_{A,B} &= t'_{A,TB} \circ Tt_{A,B} \circ \mu_{A \otimes B}.\end{aligned}$$

The monad is *commutative* if  $\ell_{A,B} = r_{A,B}$  for all objects  $A, B$ .

In probability theory,  $\ell_{A,B} = r_{A,B} : TA \otimes TB \rightarrow T(A \otimes B)$  maps two distributions to their *product distribution*.

Every monad induces a new category whose morphisms already carry the effect: a Kleisli morphism  $A \rightarrow B$  is a plain morphism  $A \rightarrow TB$ , i.e., an *effectful* process such as a Markov kernel (Definition 5). Composition uses the multiplication  $\mu$  to flatten the accumulated effect, and identities are the units  $\eta$ .

**Definition 41** (Kleisli category of a monad). Let  $(T, \eta, \mu)$  be a monad on a category  $\mathcal{C}$  (Definition 29). The *Kleisli category*  $\mathcal{C}_T$  has the following constituents.

- 1) *Objects*: the same as those of  $\mathcal{C}$ .
- 2) *Morphisms*: a Kleisli morphism  $A \rightarrow_T B$  is a  $\mathcal{C}$ -morphism  $f : A \rightarrow TB$ ; thus  $\mathcal{C}_T(A, B) = \mathcal{C}(A, TB)$ .
- 3) *Composition*: for  $f : A \rightarrow TB$  and  $g : B \rightarrow TC$ , the Kleisli composite is  $f \circ_T g = f \circ Tg \circ \mu_C : A \rightarrow TC$ .
- 4) *Identities*: the Kleisli identity on  $A$  is the unit  $\eta_A : A \rightarrow TA$ .

The monad laws of Definition 38 are exactly what make composition associative and unital, so  $\mathcal{C}_T$  is a category.

When the monad additionally interacts well with the tensor product, this effectful category inherits the same parallel-composition structure as the base category. This is the fact that underlies the compositional treatment of distributional DPs in this paper.

**Theorem 6.** The Kleisli category  $\mathcal{C}_T$  (Definition 41) of a commutative strong monad (Definition 40)  $(T, \eta, \mu, t)$  on a symmetric monoidal category  $(\mathcal{C}, \otimes, I)$  (Definition 35) is itself symmetric monoidal, with the following constituents.

- 1) The tensor product acts on objects as in  $\mathcal{C}$ , i.e.,  $A \otimes_T B = A \otimes B$ , and the unit object is the image of  $I$ .
- 2) On morphisms  $f : A \rightarrow TB$  and  $g : C \rightarrow TD$ , the Kleisli tensor  $f \otimes_T g : A \otimes C \rightarrow T(B \otimes D)$  is  $\ell_{B,D} \circ (f \otimes g)$ , using the canonical map  $\ell_{B,D} : TB \otimes TD \rightarrow T(B \otimes D)$  of Definition 40; commutativity guarantees this is well defined and functorial.
- 3) The associator, unitors, and symmetry are the images under  $\eta$  of those of  $\mathcal{C}$ .

**5) Higher and enriched structure:** Sometimes the morphisms between two objects themselves form a category (there are “morphisms between morphisms”), as with systems, the maps between them, and homotopies/2-cells between those maps. A 2-category makes this precise by adding 2-cells with two compatible ways to compose: *vertically* (within a hom-category) and *horizontally* (along objects).

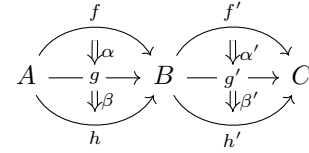
**Definition 42** (Strict and general 2-categories). A *strict 2-category*  $\mathcal{K}$  consists of the following constituents.

- 1) A collection of *0-cells* (objects)  $A, B, \dots$

- 2) For each pair  $A, B$ , a category  $\mathcal{K}(A, B)$  whose objects are *1-cells*  $f : A \rightarrow B$  and whose morphisms are *2-cells*  $\alpha : f \Rightarrow g$ .
- 3) *Vertical composition*: composition inside each  $\mathcal{K}(A, B)$ , denoted as  $\alpha \circ_V \beta$ .
- 4) *Horizontal composition*: functors  $\mathcal{K}(A, B) \times \mathcal{K}(B, C) \rightarrow \mathcal{K}(A, C)$  composing 1-cells and 2-cells along 0-cells, with identity 1-cells  $\text{id}_A$ . Denote those functors by  $\circ_H$ .

These are subject to the following conditions.

- 1) *Strict coherence*: horizontal composition is strictly associative and unital.
- 2) *Interchange law relating the two compositions*: the following two readings of a  $2 \times 2$  pasting agree:  $(\alpha \circ_H \alpha') \circ_V (\beta \circ_H \beta') = (\alpha \circ_V \beta) \circ_H (\alpha' \circ_V \beta')$ .



A *general 2-category* (or *bicategory*) weakens the coherence condition: associativity and unitality of horizontal composition hold only up to specified natural isomorphisms (associators and unitors), themselves required to satisfy coherence axioms (a pentagon and triangle) similar to Definition 34.

Finally, enrichment replaces bare hom-sets with hom-objects living in some monoidal category  $\mathcal{V}$  of “structured hom-spaces”, so that the morphisms between two objects carry extra structure (e.g., an order, a metric, or a topology) and composition respects it. Ordinary categories are the special case  $\mathcal{V} = \text{Set}$ . In this paper, enrichment of distributional uncertain DPs with QUS allows the further constructions of reparameterization as in [46].

**Definition 43** (Enriched category). Let  $\mathcal{V} = (\mathcal{V}_0, \otimes, I)$  be a monoidal category (Definition 34). A  $\mathcal{V}$ -enriched category  $\mathcal{C}$  consists of the following constituents.

- 1) A collection of objects  $A, B, C, \dots$
- 2) For each pair  $A, B$ , a *hom-object*  $\mathcal{C}(A, B) \in \text{Ob}(\mathcal{V})$  (in place of a hom-set).
- 3) For each triple  $A, B, C$ , a *composition* morphism  $c_{A,B,C} : \mathcal{C}(B, C) \otimes \mathcal{C}(A, B) \rightarrow \mathcal{C}(A, C)$  in  $\mathcal{V}$ .
- 4) For each object  $A$ , an *identity* morphism  $j_A : I \rightarrow \mathcal{C}(A, A)$  in  $\mathcal{V}$ .

These are subject to the following conditions, expressed as commuting diagrams in  $\mathcal{V}$ .

- 1) *Associativity*: composition is associative up to the associator  $\alpha$  of  $\mathcal{V}$ .

$$\begin{array}{ccc} (\mathcal{C}(C, D) \otimes \mathcal{C}(B, C)) \otimes \mathcal{C}(A, B) & \xrightarrow{\alpha} & \mathcal{C}(C, D) \otimes (\mathcal{C}(B, C) \otimes \mathcal{C}(A, B)) \\ \downarrow c \otimes \text{id} & & \downarrow \text{id} \otimes c \\ \mathcal{C}(B, D) \otimes \mathcal{C}(A, B) & & \mathcal{C}(C, D) \otimes \mathcal{C}(A, C) \\ \downarrow c & & \downarrow c \\ \mathcal{C}(A, D) & \xlongequal{\quad} & \mathcal{C}(A, D) \end{array}$$

- 2) *Unitality*: precomposing/postcomposing with  $j$  equals the

unitors  $\lambda, \rho$  of  $\mathcal{V}$ .

$$\begin{array}{ccc}
 I \otimes \mathcal{C}(A, B) & \xrightarrow{j_B \otimes \text{id}} & \mathcal{C}(B, B) \otimes \mathcal{C}(A, B) \\
 \searrow \lambda_{\mathcal{C}(A, B)} & & \downarrow c \\
 & & \mathcal{C}(A, B) \\
 \\ 
 \mathcal{C}(A, B) \otimes I & \xrightarrow{\text{id} \otimes j_A} & \mathcal{C}(A, B) \otimes \mathcal{C}(A, A) \\
 \searrow \rho_{\mathcal{C}(A, B)} & & \downarrow c \\
 & & \mathcal{C}(A, B)
 \end{array}$$

In enriched categories, the concept of “one morphism in  $\mathcal{C}(A, B)$ ” is conceptually replaced by  $\mathcal{V}$ -morphisms  $f: I \rightarrow \mathcal{C}(A, B)$ . Composing two morphisms  $f: I \rightarrow \mathcal{C}(A, B)$  and  $g: I \rightarrow \mathcal{C}(B, C)$  results in  $I \xrightarrow{\rho_I^{-1}} I \otimes I \xrightarrow{f \otimes g} \mathcal{C}(A, B) \otimes \mathcal{C}(B, C) \xrightarrow{c} \mathcal{C}(A, C)$ .

When the base  $\mathcal{V}$  is symmetric monoidal, its symmetry lets us also form tensor products of enriched categories and speak of enriched functors that respect this structure; combining the two notions gives the setting used for parametrized and higher-order co-design.

**Definition 44** (Symmetric monoidal enriched category). Let  $\mathcal{V}$  be a symmetric monoidal category (Definition 35). A *symmetric monoidal  $\mathcal{V}$ -enriched category* is a  $\mathcal{V}$ -enriched category  $\mathcal{C}$  (Definition 43) equipped with the following constituents.

- 1) *Parallel product on hom-objects*: for every  $A, B, A', B' \in \mathcal{C}$ , a  $\mathcal{V}$ -morphism

$$\otimes_{A, B, A', B'} : \mathcal{C}(A, B) \otimes \mathcal{C}(A', B') \longrightarrow \mathcal{C}(A \otimes A', B \otimes B'),$$

implementing the parallel product of hom-objects.

- 2) *Enriched associator*: for every  $A, B, C \in \mathcal{C}$ , a  $\mathcal{V}$ -isomorphism

$$\alpha_{A, B, C}^{\mathcal{C}} : I \longrightarrow \mathcal{C}((A \otimes B) \otimes C, A \otimes (B \otimes C)).$$

- 3) *Enriched unitors*: for every  $A \in \mathcal{C}$ ,  $\mathcal{V}$ -isomorphisms

$$\lambda_A^{\mathcal{C}} : I \longrightarrow \mathcal{C}(I \otimes A, A), \quad \rho_A^{\mathcal{C}} : I \longrightarrow \mathcal{C}(A \otimes I, A).$$

- 4) *Enriched symmetry*: for every  $A, B \in \mathcal{C}$ , a  $\mathcal{V}$ -isomorphism

$$\gamma_{A, B}^{\mathcal{C}} : I \longrightarrow \mathcal{C}(A \otimes B, B \otimes A).$$

These are subject to the following conditions.

- 1) *Functoriality of  $\otimes$* : the parallel product is compatible with composition,

$$\otimes_{A, C, A', C'} \circ (c \otimes c) = c \circ (\otimes_{B, C, B', C'} \otimes \otimes_{A, B, A', B'}) \circ \sigma,$$

where  $\sigma$  uses the symmetry  $\gamma$  of  $\mathcal{V}$  to interchange the middle factors; and compatible with units,  $\otimes_{A, A, A', A'} \circ (j_A \otimes j_{A'}) = j_{A \otimes A'} \circ \lambda^{-1}$ .

- 2) *Pentagon axiom*: the enriched associators satisfy the pentagon identity as an equality of  $\mathcal{V}$ -morphisms  $I \rightarrow \mathcal{C}(((A \otimes B) \otimes C) \otimes D, A \otimes (B \otimes (C \otimes D)))$ .

- 3) *Triangle axiom*: the enriched associator and unitors satisfy the triangle identity as an equality of  $\mathcal{V}$ -morphisms  $I \rightarrow \mathcal{C}((A \otimes I) \otimes B, A \otimes B)$ .

- 4) *Hexagon axiom*: the enriched symmetry and associator satisfy the hexagon identity as an equality of  $\mathcal{V}$ -morphisms  $I \rightarrow \mathcal{C}(A \otimes (B \otimes C), B \otimes (C \otimes A))$ .

- 5) *Involution*:  $\gamma_{B, A}^{\mathcal{C}}$  and  $\gamma_{A, B}^{\mathcal{C}}$  compose to the identity, i.e.,  $c \circ (\gamma_{A, B}^{\mathcal{C}} \otimes \gamma_{B, A}^{\mathcal{C}}) \circ \lambda^{-1} = j_{A \otimes B}$ .